IAT New Test 19

Question paper, answer key and worked solutions

60 questions 180 minutes 240 marks

Biology 1–15; Chemistry 16–30; Mathematics 31–45; Physics 46–60.

Correct +4; incorrect -1; unattempted 0. All 60 questions are available.

This is a provisional 2026-pattern authoring draft for 2027 preparation. Independent academic review is pending.

Instructions

Each item has exactly one correct option. Use the separate solution section for review after attempting the paper. **Figures and equations are part of the questions.**

Biology

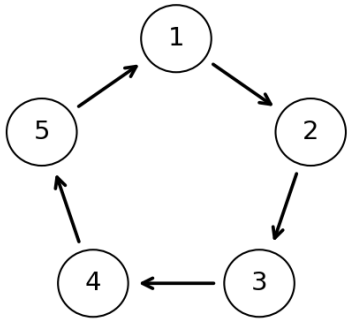
Q1 An alga contains chlorophyll a and c, much fucoxanthin, and reserves consisting mainly of laminarin and mannitol. A second sample has the same brown appearance because of an added dye, but contains chlorophyll a and b and starch. Which conclusion uses the biological evidence correctly?

A) Both must be brown algae because colour determines class
B) The first fits brown algae; the dyed second sample fits green algae
C) The first fits red algae because all accessory pigments imply that class
D) The reserve products identify both as fungi regardless of chlorophyll

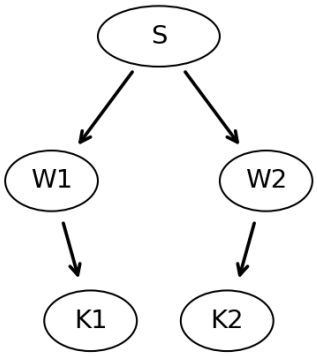
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Q2 The flower diagrams show petal overlap in an unopened bud. In X, each of five petals overlaps its next neighbour in the same direction. In Y, the large posterior petal overlaps two lateral petals, which overlap two smaller anterior petals. Which assignment is correct?

Petal-overlap maps: arrow means overlaps



X



Y

Posterior above; anterior below

A) X valvate; Y twisted
B) X vexillary; Y imbricate without a fixed large posterior petal
C) X imbricate; Y valvate
D) X twisted; Y vexillary

Q3 A brief labelled-amino-acid pulse is followed by an unlabelled chase in cells secreting protein P. P is found mainly in rough ER at 2 min, Golgi at 8 min and medium at 20 min. A drug added at 3 min leaves P stable in the ER at 8 and 20 min, while a pre-existing Golgi pool of another secretory protein reaches the medium normally. Cytosolic translation and ATP remain normal. Which localisation is best supported for the blocked step?

A) Exit from ER towards Golgi
B) All ribosomal peptide-bond formation
C) Fusion of all Golgi-derived secretory vesicles with the cell surface
D) Extracellular degradation of P

Q4 An illuminated C3 leaf receives a short pulse of labelled CO2 and then ordinary CO2. Label first appears in 3-phosphoglycerate and later in triose phosphate and regenerated RuBP. A treatment stops RuBP regeneration but leaves Rubisco functional initially. Which short-term pattern is most reasonable?
A) CO2 fixation continues indefinitely at its original rate because Rubisco remains active
B) RuBP accumulates because it is never consumed in fixation
C) Existing RuBP permits a transient fixation burst, followed by decline as acceptor is depleted
D) The first stable labelled product must change from a three-carbon to a six-carbon compound

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Q5 After surgical removal of the thyroid gland, a patient receives sufficient thyroxine replacement to keep circulating thyroxine normal. The hypothalamus and pituitary are intact. Which prediction follows from negative feedback?
A) TSH must rise without bound because the gland is absent
B) TSH can remain near its feedback-regulated range despite the absent gland
C) Replacement thyroxine directly becomes TSH
D) TSH must be zero because all endocrine feedback is irreversible

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Q6 An unfertilised ovary is induced to form a fruit by a hormone treatment, while its ovules do not develop embryos. In a different plant, seeds form embryos without fertilisation. Which pair names these two phenomena, respectively?
A) Parthenocarpy and apomixis
B) Polyembryony and double fertilisation
C) Self-incompatibility and apomixis
D) Apomixis and parthenocarpy

Q7 A fully penetrant autosomal-recessive condition is caused by a. Two unaffected adults are independently known to be Aa; one previous child is aa. With independent meioses and no survival difference, what is the probability that their next two children consist of exactly one affected child and one unaffected carrier, in either order?
A) 1/4
B) 1/8
C) 3/8
D) 1/2

Q8 An infant receives maternal antibodies and later an antigen-specific vaccine. The maternal antibodies gradually disappear. Months afterward, when no transferred antibody is detectable, the infant gives a faster response to the same vaccine antigen than an unvaccinated matched control. Which interpretation fits both time periods?
A) Maternal antibodies are continually replicated by the infant's plasma
B) Both protections must be passive because the recipient is an infant
C) Early passive protection and later vaccine-induced immunological memory are distinguishable
D) Loss of maternal antibody means the vaccine could not have acted

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Q9 Two matched bacterial clones contain the same inducible foreign gene and grow equally under selection. X makes the target RNA after induction but no target protein; Y makes both. DNA sequencing confirms identical promoters and protein-coding sequences, but X has a change in its ribosome-binding region. Which next test most directly addresses the proposed failure in X?
A) Repeat only the antibiotic-survival test
B) Measure plasmid colour without induction
C) Remove the promoter from Y and compare survival
D) Restore X's ribosome-binding sequence and test protein recovery while monitoring RNA

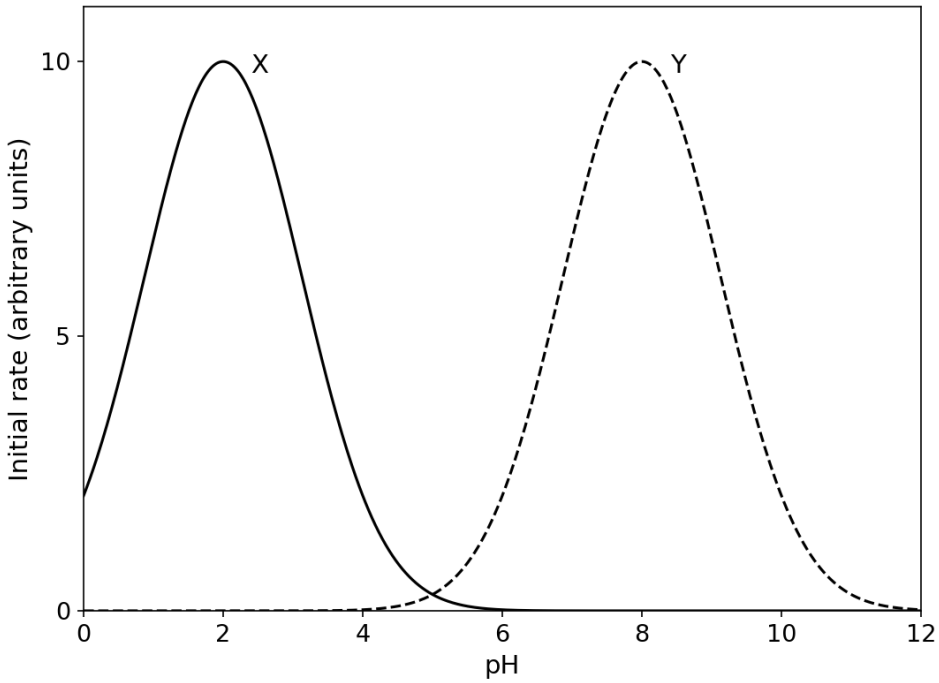
Q10 A food web contains phytoplankton with small standing biomass and rapidly increasing biomass over repeated growth-and-consumption cycles. Fish have larger standing biomass but depend on the phytoplankton production. Which ecological pyramid can be inverted here without implying an upward gain of energy?
A) The energy pyramid, because fish are larger
B) The biomass pyramid, because producer turnover can replenish a small standing stock
C) Both pyramids, because standing biomass measures annual energy transfer
D) Neither pyramid, because every biomass pyramid is upright

Q11 A male with phase AB/ab and a female with phase Ab/aB mate. Recombination fraction is 0.20 in each sex, meioses are independent, and all offspring survive. What fraction of offspring is aabb?
A) 1/100
B) 4/25
C) 1/25
D) 1/4

Q12 A woman heterozygous for an X-linked recessive allele mates with an unaffected man. Assume equal sex probabilities, full penetrance and independent births. Given that neither of their two children is affected, what is the probability that both are carrier daughters?
A) 1/9
B) 1/16
C) 1/4
D) 4/9

Q13 During shortening, one sarcomere changes length from 2.4 micrometres to 2.0 micrometres. Its thick filaments remain 1.6 micrometres long and each thin filament remains 1.0 micrometre long. Assume symmetric overlap with thin filaments extending inward from both Z lines. What happens to the H-zone width?
A) It grows from 0.4 to 0.8 micrometre
B) It remains 1.6 micrometres
C) It shrinks from 0.8 to 0.4 micrometre
D) It shrinks from 0.4 micrometre to zero

Q14 The figure shows initial enzyme rate versus pH for equal enzyme amounts. X peaks near pH 2 and Y near pH 8. Rates are assayed separately with each enzyme's own substrate at saturating concentration. Which inference is justified?



A) X must have a smaller molecular mass than Y
B) The data support different optimal protonation environments but do not rank their substrate affinities
C) Y must be a competitive inhibitor of X
D) X can never regain activity after any departure from pH 2

Q15 For comparable habitat, species richness follows $S = cA^z$. A plot of area A has 16 species and one of area 4A has 32. Two separate 4A patches each have 32 species and share k species. Assume the same fitted law applies to a single contiguous 8A patch. For the two-patch union to contain more species than that single patch, what exact condition on k is required?
A) $k > 64 - 32\sqrt{2}$
B) $k < 32\sqrt{2}$
C) $k < 64 - 32\sqrt{2}$
D) $k = 0$ only

Chemistry

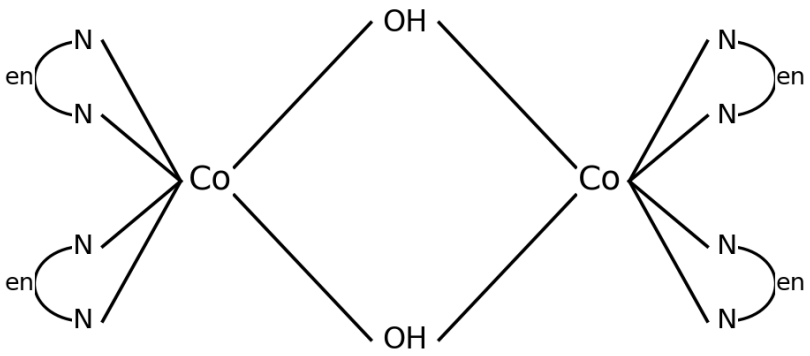
Q16 An amide's C-N unit is nearly planar and has a larger rotational barrier than the C-N bond of an ordinary amine. A resonance contributor places a C=N bond, positive charge on N and negative charge on O. Which interpretation connects these observations?

A) The amide rapidly alternates between two separately isolable charged molecules
B) The amide nitrogen must have five covalent bonds in its ground state
C) Electron delocalisation gives partial C-N double-bond character, so twisting weakens conjugation
D) The C-N barrier proves the carbonyl oxygen is not bonded to carbon

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Q17 The schematic represents $[Co_2(en)_4(OH)_2]^{4+}$. Each en chelates one cobalt through two N atoms; each of the two OH groups bridges both cobalt centres through its O atom. All other donor connections are shown. What are the oxidation state and coordination number of each equivalent cobalt?

Bridged coordination ion: overall charge 4+

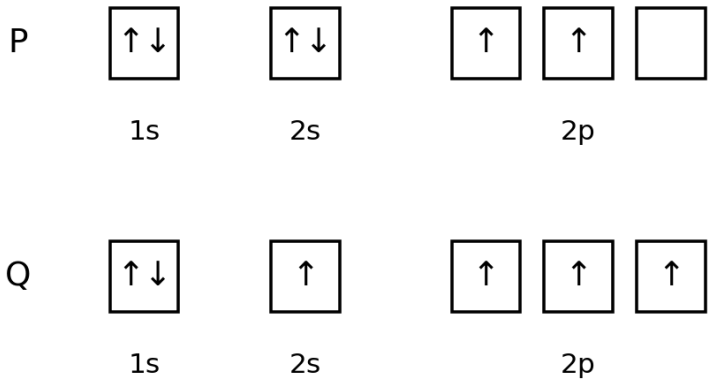


en neutral; each OH has charge −1

A) +3 and 6
B) +2 and 6
C) +3 and 4
D) +4 and 8

Q18 The figure gives two allowed carbon-atom configurations: P is 1s2 2s2 2p2 with the p electrons unpaired in different orbitals; Q is 1s2 2s1 2p3 with parallel p spins. Ignoring orbital contributions to magnetism, which comparison is correct?

Carbon configurations P and Q



A) Q is forbidden because a 2s electron cannot be promoted
B) P and Q have equal energy because electron number is equal
C) P has four unpaired electrons and Q has two
D) P has two unpaired electrons; Q has four and is excited relative to P

Q19 In acid-catalysed addition to a carbonyl compound, the carbonyl oxygen first uses a lone pair to bind H+. Which description of that first step and its consequence is correct?
A) Carbonyl carbon donates a pair to H+, making it less electrophilic
B) Oxygen acts as a nucleophile towards H+, and protonation increases carbonyl-carbon electrophilicity
C) H+ donates an electron pair to oxygen and becomes a radical
D) Protonation removes the carbonyl carbon from the molecule

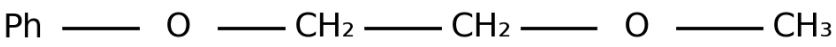
Q20 Toluene is chlorinated in separate trials. X uses light and conditions favouring one side-chain substitution. Y uses Cl2/FeCl3 in the dark under ring-substitution conditions. Which products best distinguish the two pathways?
A) X mainly o/p-chlorotoluene; Y benzyl chloride
B) Both mainly benzyl chloride because methyl always reacts first
C) X mainly benzyl chloride; Y mainly o/p-chlorotoluene
D) Both mainly m-chlorotoluene because chlorine is deactivating

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Q21 A student proposes preparing pure propane by the Wurtz reaction of an equimolar mixture of methyl bromide and ethyl bromide with sodium in dry ether. Assuming both self-coupling and cross-coupling occur, why is the proposal unsuitable for a single pure product?
A) Ethane, propane and butane can all form
B) Only methane can form because sodium removes carbon
C) Propane cannot form because unlike alkyl groups never couple
D) All coupling products must have an odd number of carbons

Q22 The structure is $C_6H_5{-}O{-}CH_2{-}CH_2{-}O{-}CH_3$. It is heated with excess HI until ether cleavage and conversion of the resulting ordinary aliphatic alcohol groups to iodides are complete. Assume substitution without elimination or skeletal rearrangement; the aryl-O bond is retained. Which final organic product set and minimum HI consumption per mole of starting ether are correct?

Starting ether connectivity



Ph = C_6H_5

A) Iodobenzene, ethane-1,2-diol and methane; 2 mol HI
B) Phenol, 2-iodoethanol and methyl iodide; 2 mol HI
C) Phenol, ethane and methyl iodide; 3 mol HI
D) Phenol, 1,2-diiodoethane and methyl iodide; 3 mol HI

Q23 An ester has formula C4H8O2. Complete hydrolysis produces a carboxylic acid and an alcohol, which are separated. The alcohol gives a positive iodoform test under standard conditions. Which ester is consistent?
A) Methyl propanoate
B) Propyl methanoate
C) Ethyl ethanoate
D) Methyl ethanoate

Q24 Which sequence converts aniline to benzoic acid while increasing the organic carbon count by one, without retaining nitrogen in the final product?
A) Diazotisation at 0-5 C, CuCN substitution, then nitrile hydrolysis
B) Diazotisation, replacement by H, then amide hydrolysis
C) Acetylation followed by Hofmann degradation of the ring
D) Bromination followed only by reduction of the C-Br bond

Q25 A dilute ideal electrolyte has measured van't Hoff factor i=5. Assume complete dissociation into simple ions, with no association or reaction. Its ions and charges are otherwise unknown. Which conclusion is justified from this measurement alone?
A) The formula must be A2B3 and no other stoichiometry is possible
B) Five ionic particles form per formula unit, but the formula is not uniquely determined
C) Each ion must carry charge magnitude five
D) The formula must contain exactly five different elements

Q26 A conductivity cell gives resistance 100 ohm for a reference solution with conductivity 0.0140 S/cm. In the same cell, a monoprotic weak acid of concentration 0.0100 mol/L has resistance 1400 ohm. Its limiting molar conductivity is 400 S cm2/mol. Using an ideal dissociation model, what is Ka?
A) 2.5×10^{-3} M
B) 6.25×10^{-4} M
C) 8.33×10^{-4} M
D) 1.0×10^{-2} M

Q27 Two independent first-order reactions X and Y have temperature-independent Arrhenius parameters. At 300 K, kX/kY=4; at 600 K, kX/kY=1. At 400 K, their initial reactant concentrations obey [X]0/[Y]0=3. What is the ratio of their initial disappearance rates at 400 K?
A) 3
B) 6
C) 2
D) 12

Q28 At a fixed temperature, K1=4 for A $\rightleftharpoons$ B and K2=8 for A $\rightleftharpoons$ C, using dimensionless activities. What are K and the sign of the standard Gibbs change for 2 B $\rightleftharpoons$ A + C?
A) K=2; negative
B) K=32; negative
C) K=1/2; negative
D) K=1/2; positive

Q29 At 25 C, a monoprotic acid has analytical concentration 10^-8 M and Ka=10^-5. Water has Kw=10^-14. With no other solutes, which pH description is correct?
A) Slightly below 7, with water autoionisation important
B) Exactly 8 because -log of the acid concentration is 8
C) Exactly 6.5 from the small-dissociation formula
D) Exactly 5 because pH always equals pKa

Q30 Zr and Hf commonly form +4 compounds with very similar ionic sizes, although Hf has a much larger atomic mass. Which explanation and inference are most appropriate?
A) Mass alone forces Hf compounds to have a different valence from Zr compounds
B) The two elements must have identical densities, isotope masses and all spectra
C) The similarity proves Hf has no f electrons in any preceding element
D) The intervening lanthanoid contraction helps give similar sizes and chemistry; equal mass-related properties do not follow

Mathematics

Q31 An ellipse has foci (-3,0) and (3,0) and major-axis length 10. A point P on its upper half is 4 units from the left focus. What is the area of the triangle formed by P and the two foci?
A) $8\sqrt{2}$
B) $4\sqrt{2}$
C) 10
D) $16\sqrt{2}$

Q32 For $f(x) = |x - \sin x|$, which statement about x=0 is correct?
A) f is not differentiable at zero
B) f is differentiable but not twice differentiable at zero
C) f is twice differentiable with f”(0)=1
D) f is twice differentiable with f”(0)=0

Q33 For real a, let $I(a) = \int_{-1}^{1} |x - a|\, dx$. What is the product of all real a satisfying I(a)=3/2?
A) -9/4
B) -1/2
C) 1/2
D) 0

Q34 Let $f : \{-2,-1,0,1,2\} \to \{0,1,4\}$ be f(x)=x squared. How many functions $g : \{0,1,4\} \to \{-2,-1,0,1,2\}$ satisfy $f(g(y)) = y$ for every y in $\{0,1,4\}$?
A) 2
B) 8
C) 4
D) 0

Q35 A real matrix has form $A = \begin{pmatrix} a & b & 0 \\ b & a & 0 \\ 0 & 0 & 2 \end{pmatrix}$ and satisfies adj $A = 2A$. What are all possible values of its trace?
A) -2 and 6
B) 2 and 6
C) -6 and 6
D) 6 only

Q36 Independent events A and B have probabilities 0.4 and 0.5. Given that exactly one of them occurs, what is the probability that A occurs?
A) 1/2
B) 2/5
C) 1/5
D) 4/5

Q37 A path from (0,0) to (4,3) uses four right steps R=(1,0) and three up steps U=(0,1). No two U steps may be consecutive, and the path must avoid vertex (2,1). How many paths satisfy both conditions?
A) 10
B) 7
C) 4
D) 3

Q38 A positive integer n satisfies $\sum_{k=0}^{n} \frac{(-1)^k}{k+1}\binom{n}{k} = \frac{1}{9}$. What is the middle binomial coefficient in $(1+x)^n$?
A) 70
B) 56
C) 126
D) 28

Q39 A sequence has partial sums $S_n = \sum_{j=1}^{n} a_j$ and satisfies $S_n = 3a_n - 2n$ for every positive integer n. Which formula gives a_n?
A) $3(3/2)^{n-1} + 2$
B) $(3/2)^{n-1}$
C) $3(3/2)^{n-1} - 2$
D) $3(2/3)^{n-1} - 2$

Q40 A function f on the real line is continuous at zero, has f(0)=1 and satisfies $f(2x) - f(x) = 2x$ for every real x. What is f'(0)?
A) 0
B) 1
C) 4
D) 2

Q41 For x>0, y satisfies $xy' = y - x$ and y(1)=1. At which positive x does y first become zero?
A) 1
B) e
C) e^{-1}
D) e^2

Q42 Reflect P=(2,1,0) across the line $\mathbf{r} = (1,0,1) + t(1,2,2)$, meaning the half-turn about this line. If the reflected point is P', what is the sum of the coordinates of P'?
A) 17/9
B) 1/9
C) 3
D) 19/9

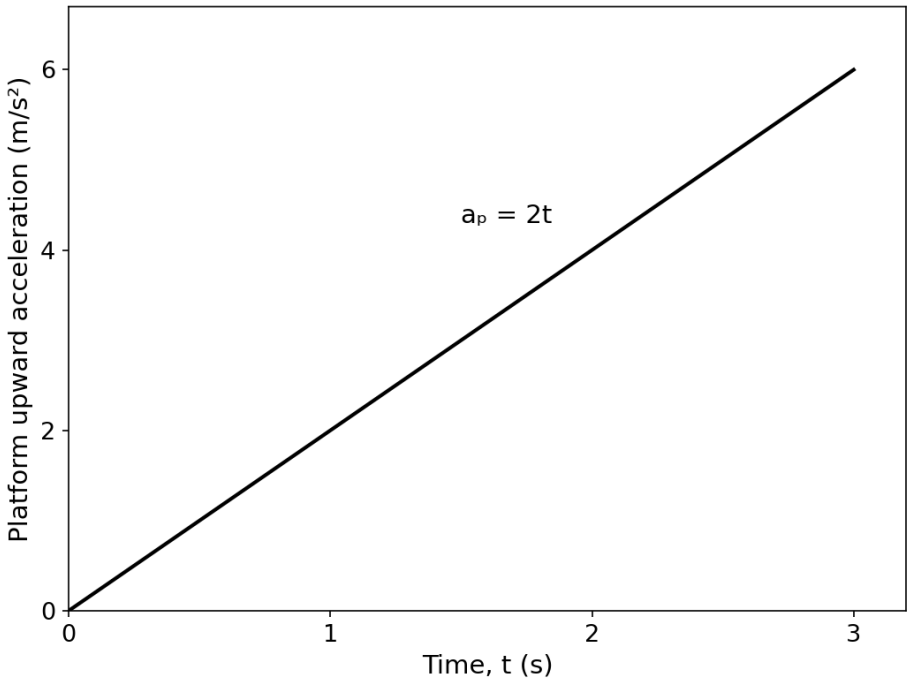
Q43 Vectors a,b,c satisfy $\mathbf{a} \times \mathbf{b} = \mathbf{a} \times \mathbf{c}$. What must be the volume of the parallelepiped formed by these three vectors?
A) $|\mathbf{a}|\,|\mathbf{b} \times \mathbf{c}|$
B) 1
C) 0
D) It can be any positive value

Q44 A discrete data table has values 0, 2 and 6 with frequencies p, 4 and 2, respectively, where p is a nonnegative integer. Its mean absolute deviation about its arithmetic mean is 5/3. What is p?
A) 6
B) 3
C) 4
D) 12

Q45 Angles alpha and beta are acute and satisfy $\tan\alpha + \tan\beta = 3$ and $\tan\alpha\tan\beta = 1/2$. What is $\cos(\alpha - \beta)$?
A) $1/\sqrt{37}$
B) $3/\sqrt{37}$
C) $3/\sqrt{7}$
D) $2/\sqrt{37}$

Physics

Q46 An open platform starts from rest at t=0 and accelerates vertically upward as shown, with $a_p(t) = 2t$ m/s2 for t in seconds. A ball is launched upward from it at t=0 with speed 18 m/s relative to the platform. Take g=10 m/s2 and ignore air resistance. At its first return to the platform, what is the ball's velocity relative to it? Upward is positive.



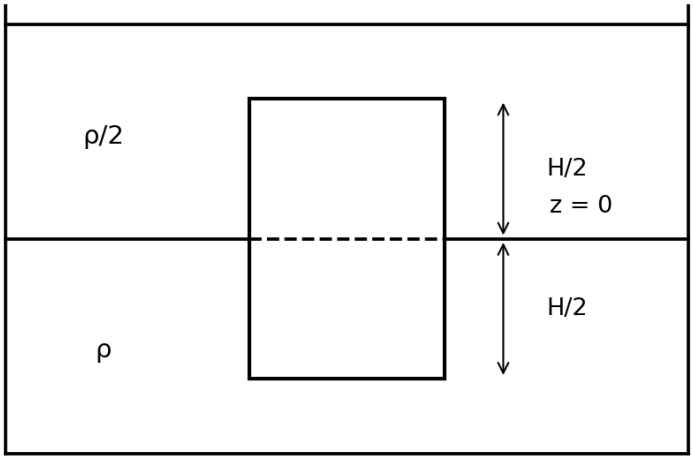
A) -12 m/s
B) -21 m/s
C) -9 m/s
D) +21 m/s

Q47 A block rests on top of a massless vertical spring and is released from rest while the spring is compressed by $3mg/(2k)$. The spring can push but cannot pull the block. Neglect damping. How far does the block rise before its first stop, and is it still in contact then?
A) $3mg/(2k)$, no
B) $9mg/(8k)$, no
C) mg/k, yes
D) $mg/(2k)$, yes

Q48 A rigid body rolls without slipping with centre-of-mass speed v. Its translational kinetic energy is 3/5 of its total kinetic energy. What is its moment of inertia about the rolling axis through its centre, if mass is M and rolling radius is R?
A) $3MR^2/2$
B) $3MR^2/5$
C) $2MR^2/5$
D) $2MR^2/3$

Q49 A uniform block of height H and constant cross-section floats fully submerged across two immiscible layers as shown: half its volume is in lower liquid of density rho and half in upper liquid of density rho/2. The upper layer is carefully removed; the large lower reservoir's surface stays at the old interface height. The block then floats in the lower liquid alone. What is its vertical displacement from the original position?

Initial block position across two liquid layers

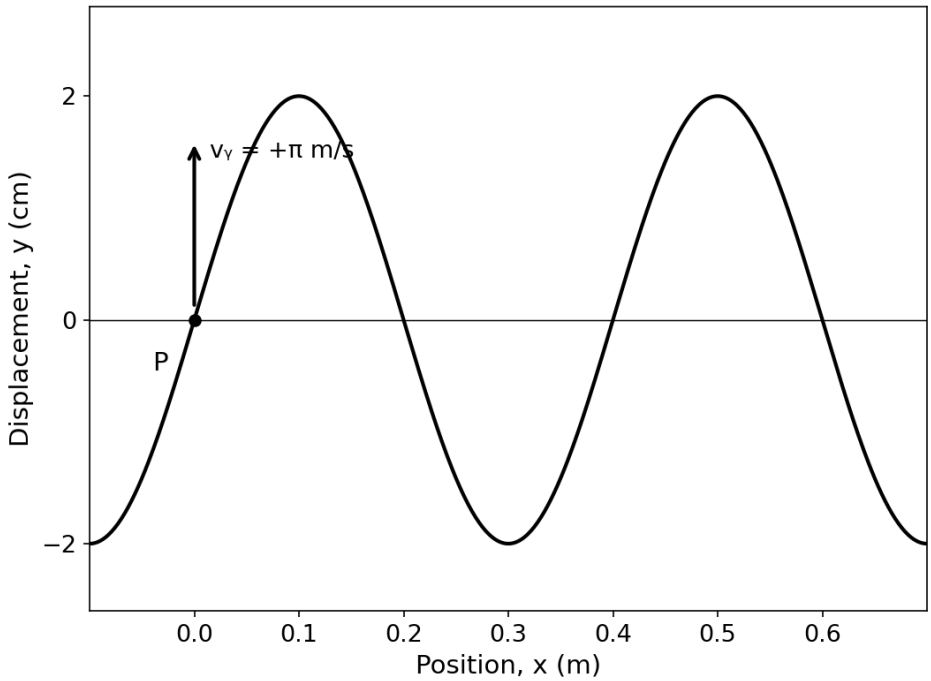


After removal: lower free surface stays at z = 0

A) Downward by H/4
B) Upward by H/4
C) Downward by H/2
D) No displacement

Q50 An ideal monatomic gas initially has pressure 3P and volume V. In an insulated cylinder, it is suddenly released to expand against constant external pressure P. It eventually settles at pressure P, with negligible piston kinetic energy at both endpoints and no heat leakage. What is the final volume?
A) $3V$
B) $3^{3/5}V$
C) $11V/5$
D) $5V/3$

Q51 The figure is a sinusoidal travelling-wave snapshot at t=0, with amplitude 2.0 cm and wavelength 0.40 m. At x=0 the string crosses equilibrium with positive spatial slope, and its instantaneous transverse velocity is +pi m/s. Which propagation direction and wave speed follow?

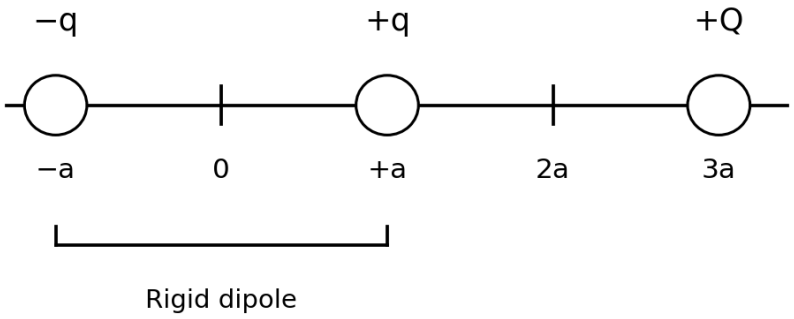


A) Rightward at 10 m/s
B) Leftward at 10 m/s
C) Leftward at 25 m/s
D) Rightward at 25 m/s

Q52 Two point masses orbit their common centre of mass in circles with period T and fixed separation d, under only mutual gravitation. Their orbital radii are d/3 and 2d/3. What is the mass of the object in the smaller orbit?
A) $4\pi^2 d^3/(3GT^2)$
B) $4\pi^2 d^3/(GT^2)$
C) $8\pi^2 d^3/(GT^2)$
D) $8\pi^2 d^3/(3GT^2)$

Q53 Charges -q and +q form a rigid dipole at x=-a and x=+a, respectively. An external fixed charge +Q is at x=3a, as shown. Ignore the dipole's self-energy. What are the exact net x force on the dipole, its torque about its centre and its external interaction energy?

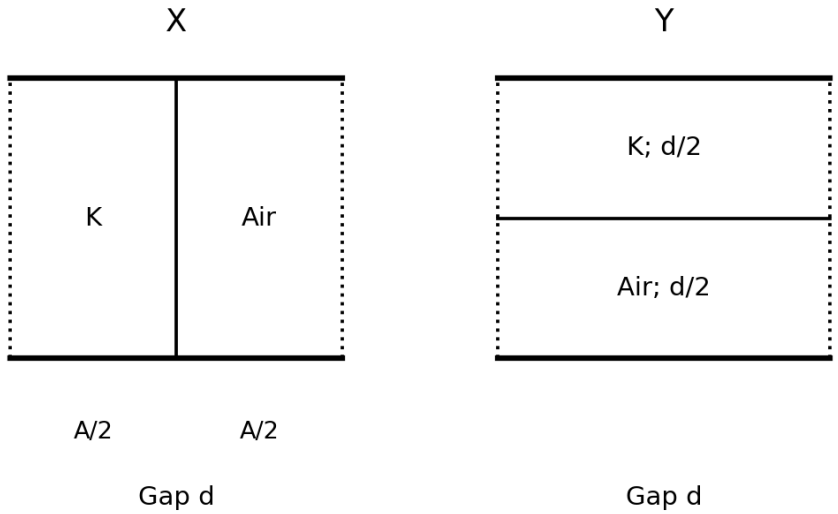
Finite dipole and external source



A) $+3kQq/(16a^2), 0, kQq/(4a)$
B) $-kQq/(9a^2), 0, 2kQq/(9a)$
C) $-3kQq/(16a^2), 0, kQq/(4a)$
D) $0, kQq/a, kQq/(4a)$

Q54 Two otherwise identical parallel-plate capacitors each have plate area A and gap d. In X, dielectric K fills half the plate area across the full gap; the rest is air. In Y, dielectric K fills half the gap thickness across the full area; the rest is air. Fringing is neglected. Their capacitances satisfy CX/CY=4/3. For K>1, what are K and UX/UY when the capacitors separately hold the same charge Q?

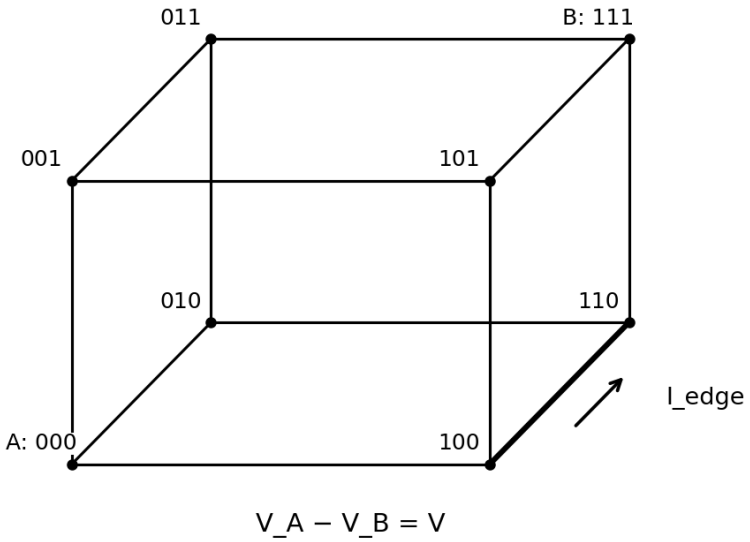
Two dielectric arrangements; same plate area A



A) $3, 3/4$
B) $3, 4/3$
C) $1/3, 3/4$
D) $4, 1/2$

Q55 Each of the 12 edges of the cube is a resistor R. An ideal source maintains voltage V between opposite vertices A and B. The figure marks one edge joining a vertex adjacent to A to a vertex adjacent to B. What are the equivalent resistance between A and B and the current magnitude in that marked edge?

Cube network: every edge is a resistor R



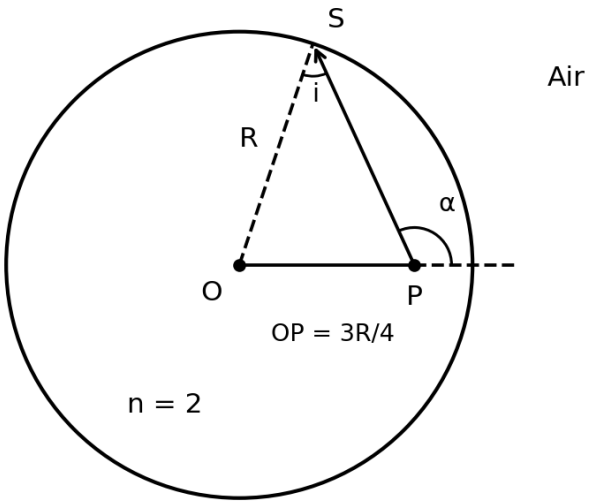
A) $R, V/(6R)$
B) $R/2, V/(3R)$
C) $5R/6, V/(6R)$
D) $5R/6, V/(5R)$

Q56 Three long parallel wires lie at transverse coordinates A=(0,0), B=(d,0), C=(d/2,sqrt3 d/2). A and B carry current I out of the coordinate plane; C carries I into it. What is the force per unit length on A?
A) $\sqrt{3}\mu_0 I^2/(2\pi d)$, towards the triangle centre
B) $\mu_0 I^2/(2\pi d)$, parallel to C towards B
C) 0
D) $2\mu_0 I^2/(2\pi d)$, towards B

Q57 An ideal transformer has primary-to-secondary turns ratio 3:1 and is supplied at 120 V rms. Its secondary drives a series RLC circuit with R=6 ohm, XL=10 ohm and XC=2 ohm at that frequency. What are the primary rms current and the power factor seen by the source?
A) 4 A, 0.6
B) $4/3$ A, 1
C) 12 A, 0.8
D) $4/3$ A, 0.6

Q58 A point source P lies inside a transparent sphere of radius R and refractive index 2, with OP=3R/4. The surrounding medium is air. Let alpha in [0,pi] be the angle between a ray's initial direction and the outward radial direction OP. Excluding rays exactly at critical incidence, which complete set reaches the surface with incidence below the critical angle on its first encounter? Let $\theta = \arcsin(2/3)$.

Ray from an off-centre source in a sphere



A) $[0, \theta) \cup (\pi - \theta, \pi]$
B) $[0, \theta)$ only
C) $(\theta, \pi - \theta)$
D) $[0, \pi]$

Q59 An ideal circular parallel-plate capacitor of plate radius a charges with a spatially uniform increasing electric field E(t). Ignore edge effects. Compare magnetic-field magnitudes in the gap at radii a/2 and 2a, using a circular Ampere-Maxwell loop and including all displacement flux only through the plate area. Which statement is correct?
A) The field at a/2 is four times that at 2a; conduction crosses the gap
B) The field at 2a is four times that at a/2; displacement current is zero
C) The fields are equal; no conduction current crosses the ideal gap
D) Both fields are zero because no electrons cross the gap

Q60 A source initially gives an expected detected signal of 80 counts/min above a constant background of 20 counts/min. After exactly one source half-life, a geometry change doubles detection efficiency without changing the source or background. What expected total count rate is then measured?
A) 60 counts/min
B) 100 counts/min
C) 180 counts/min
D) 50 counts/min

Answer key and worked solutions

Every solution explains the selected choice and the alternative errors.

Biology

Q1 / B Diagnostic pigments and reserves despite an artificial colour change

Answer: (B) The first fits brown algae and the second green algae.

Why: Natural pigment complement and reserve chemistry provide linked classification characters. Fucoxanthin with chlorophyll a and c and laminarin/mannitol reserves is characteristic of brown algae. Chlorophyll a and b with starch supports green algae despite an artificial change of visible colour. A treatment that changes appearance does not automatically replace the underlying biochemical traits used for identification.

Why not the others: (A) treats an experimentally altered surface observation as decisive taxonomy. (C) ignores the specific pigment and reserve combination associated with brown algae. (D) ignores photosynthetic pigment evidence and misassigns algal reserves to a fungal diagnosis.

Key point: Classification should combine stable diagnostic characters rather than a manipulated appearance.

Q2 / D Cyclic petal overlap versus a standard-wing-keel hierarchy

Answer: (D) X twisted and Y vexillary.

Why: X has a repeated directional overlap around the entire whorl, the defining arrangement of twisted aestivation. Y has a standard-wing-keel hierarchy: the large posterior standard overlaps the wings, which overlap the anterior keel petals. That ordered special imbricate arrangement is vexillary aestivation. The relative overlap, rather than whether the petals are fused, supplies the evidence.

Why not the others: (A) calls an overlapping whorl valvate, although valvate margins meet without overlap. (B) assigns the standard-wing-keel pattern to X even though X has uniform cyclic overlap. (C) misses the specific twisted pattern in X and denies the clear overlap in Y.

Key point: Trace who overlaps whom around the bud before naming aestivation.

Q3 / A Pulse-chase localisation with a downstream secretory bypass

Answer: (A) ER-to-Golgi transfer is blocked.

Why: The untreated time course follows the secretory route from rough ER through Golgi to the external medium. With drug, an already synthesised stable cohort remains in the first compartment instead of appearing downstream. Normal export of a pre-existing Golgi pool shows that downstream secretion can still operate, while maintained ATP and cytosolic translation argue against a general energy or translation collapse. Together the tracer position and bypass pool localise the earliest affected transfer to ER exit or delivery towards Golgi, without identifying a particular molecular transport protein.

Why not the others: (B) contradicts preserved translation and does not explain stable accumulation of the pre-existing labelled cohort in ER. (C) predicts accumulation of the bypass pool downstream as well, contrary to its secretion. (D) would act after export and cannot explain the persistent intracellular ER localisation.

Key point: Time-resolved localisation plus a downstream bypass pool can identify a blocked trafficking stage.

Q4 / C Transient fixation after interruption of Calvin-cycle acceptor regeneration

Answer: (C) Fixation is transiently maintained and then declines.

Why: Rubisco can use the RuBP pool already present when regeneration stops. Each fixation consumes an acceptor molecule, however, and the pool cannot sustain continued flux without its replenishing reactions. The observed order of label supports movement through a cycle whose acceptor is regenerated, rather than a single irreversible store of unlimited capacity. The treatment changes the supply of RuBP, not the identity of the initial stable C3 product.

Why not the others: (A) treats enzyme activity as sufficient despite depletion of its substrate. (B) reverses the pool balance after its production is stopped but consumption continues. (D) changes the primary fixation chemistry without a stated change in pathway.

Key point: An intact enzyme cannot sustain pathway flux when a required regenerated acceptor is exhausted.

Q5 / B Hormone replacement restores an upstream feedback signal

Answer: (B) TSH can remain near its feedback-regulated range.

Why: The feedback signal received by the hypothalamic-pituitary axis is circulating thyroid hormone. Adequate replacement can provide that signal even though the thyroid tissue no longer produces it. The missing gland prevents a normal thyroid response to TSH, but does not prevent replacement hormone from suppressing excessive upstream stimulation. Actual dosing and secretion vary, so the prediction concerns the direction of regulation rather than an exact numerical TSH.

Why not the others: (A) ignores the restored circulating feedback signal. (C) mistakes regulation of one hormone's secretion for chemical conversion into another. (D) imposes complete irreversible shutdown rather than adjustable feedback.

Key point: Feedback depends on the circulating signal, not solely on the anatomical presence of its usual source.

Q6 / A Unfertilised fruit formation contrasted with asexual seed formation

Answer: (A) Parthenocarpy and apomixis.

Why: The first observation concerns development of the fruit without fertilisation, which is parthenocarpy. It need not imply embryo development or a viable seed. The second concerns asexual seed/embryo formation, the defining distinction for apomixis. Naming the developing structure prevents confusion between an ovary-derived fruit and an ovule-associated seed.

Why not the others: (D) reverses fruit formation and asexual seed formation. (B) requires multiple embryos or ordinary gametic fusion, neither of which is stated. (C) describes rejection of self-pollen rather than development of an unfertilised ovary into fruit.

Key point: Ask whether the observation is about fruit development or embryo formation.

Q7 / A Ordered sibling outcomes with known carriers rather than inferred parental risk

Answer: (A) 1/4.

Given: Parents are known $Aa \times Aa$; previous birth does not change their genotypes.

Formula/Principle: Use mutually exclusive ordered outcomes for independent births.

Step 1: For each birth,

$$P(aa) = \tfrac{1}{4}, \qquad P(Aa) = \tfrac{1}{2}.$$

Step 2: The required outcomes are aa then Aa, or Aa then aa:

$$P = \tfrac{1}{4}\tfrac{1}{2} + \tfrac{1}{2}\tfrac{1}{4} = \tfrac{1}{4}.$$

Why: Carrier status is narrower than merely unaffected status. The two orders are distinct and exhaustive for the requested event. A previous affected child is consistent with the already known cross and does not impose a quota on later Mendelian outcomes.

Why not the others: (B) counts only one birth order. (C) uses all unaffected children, including

AA, in place of the required carrier. (D) uses one child's carrier probability without requiring the other to be affected.

Key point: Specify both genotype classes and all admissible birth orders before multiplying.

Q8 / C Transferred antibody decay separated from vaccine-generated recall

Answer: (C) Early passive protection differs from later active memory.

Why: Transferred maternal antibodies can provide immediate protection without the infant having generated those molecules or the corresponding memory. Their decline is compatible with turnover of a finite transferred pool. A later enhanced antigen-specific response after that pool has disappeared is evidence of an acquired response established by the infant's own immune system after vaccination. The observations therefore separate the source of early effectors from the source of later recall.

Why not the others: (A) treats antibodies as self-replicating genetic templates. (B) assigns immune mechanism by age rather than by whether the recipient generated a response. (D) confuses disappearance of transferred effectors with absence of actively acquired memory.

Key point: Follow the origin and persistence of both effectors and memory when classifying immunity.

Q9 / D Target RNA without protein and a translation-site rescue

Answer: (D) Restore the binding sequence and measure both RNA and protein.

Why: Target RNA in X shows that transcription has occurred, while absent protein places the unresolved failure downstream of that observation. A change in the region needed for ribosome recruitment offers a translation-level hypothesis. Restoring that specific sequence is a causal rescue test; monitoring RNA checks that protein recovery is not merely the result of a new increase in transcription. Equal growth and identical coding sequence help narrow the interpretation without proving every alternative has vanished.

Why not the others: (A) measures a selectable marker's function, not the foreign gene's translation. (B) has no specified connection to target RNA or protein. (C) destroys transcription in the working clone and does not repair the candidate defect in X.

Key point: A selectable clone need not express the desired protein; test the stage indicated by RNA and protein evidence.

Q10 / B Inverted standing biomass with an upright energy flow

Answer: (B) The biomass pyramid can be inverted.

Why: Standing biomass is a stock measured at a time, whereas production and energy transfer are flows over an interval. Rapidly turning-over phytoplankton can supply consumers repeatedly while their instantaneous stock stays small. That allows the aquatic biomass pyramid to be inverted. Energy transfer still loses usable energy between trophic levels, so the energy pyramid remains upright for a consistently measured system.

Why not the others: (A) confuses organism size with trophic energy supply. (C) treats a stock as though it were a time-integrated flow. (D) ignores turnover-supported inverted biomass pyramids in aquatic systems.

Key point: Separate standing stocks from production rates before interpreting a pyramid.

Q11 / C Opposite linkage phases in the two parents of an intercross

Answer: (C) 1/25.

Given: Male coupling phase and female repulsion phase have $r = 0.20$.

Formula/Principle: Offspring aabb requires an ab gamete from each parent; identify whether that gamete is parental in each.

Step 1: In AB/ab, ab is parental, so

$$P_m(ab) = (1 - r)/2 = 0.40.$$

Step 2: In Ab/aB, ab is recombinant, so

$$P_f(ab) = r/2 = 0.10.$$

Step 3:

$$P(aabb) = 0.40(0.10) = 0.04 = 1/25.$$

Why: Equal recombination fractions do not give equal probabilities to every named gamete when parental phases differ. The required recessive genotype selects one particular gamete in each parent. Phase must therefore be assigned before applying multiplication across independent meioses.

Why not the others: (A) treats ab as recombinant in both parents. (B) treats it as parental in both. (D) ignores linkage and uses an unrelated single-locus recessive probability.

Key point: The same named gamete can be parental in one parent and recombinant in the other.

Q12 / A Carrier-daughter pair conditional on no affected child

Answer: (A) 1/9.

Given: Per birth, carrier daughter, noncarrier daughter, unaffected son and affected son each have probability $1/4$.

Formula/Principle: Condition the two-birth event on the exclusion of affected children.

Step 1:

$$P(\text{two carrier daughters}) = (1/4)^2 = 1/16.$$

Step 2: Each child is unaffected with probability $3/4$, so

$$P(\text{neither affected}) = (3/4)^2 = 9/16.$$

Step 3:

$$P = \frac{1/16}{9/16} = 1/9.$$

Why: The condition removes affected sons but retains two different daughter classes and unaffected sons. It does not assert that both children are girls. The numerator is already contained in the allowed event, which makes direct conditional division appropriate.

Why not the others: (B) omits the stated conditioning. (C) conditions on both children being daughters instead. (D) counts all daughter pairs rather than carrier daughters only.

Key point: Translate a family-level condition into its exact retained outcome set.

Q13 / D H-zone disappearance from explicit filament-tip geometry

Answer: (D) It shrinks from 0.4 micrometre to zero.

Given: Thin filaments extend 1.0 micrometre from each end; sarcomere length changes from 2.4 to 2.0 micrometres.

Formula/Principle: The H zone is the central thick-filament segment not overlapped by thin filaments.

Step 1: The separation of the inward thin-filament tips is

$$H = \max(0, L - 2\ell_{\text{thin}}).$$

Step 2: Initially,

$$H_i = 2.4 - 2(1.0) = 0.4 \text{ micrometre}.$$

Step 3: Finally,

$$H_f = 2.0 - 2(1.0) = 0.$$

Why: Sliding increases overlap without shortening either filament type. The H-zone boundary is set by the inward tips of the thin filaments, not by the thick filament's total length. In this stated symmetric geometry the tips just meet at the final length, eliminating the unoverlapped central region.

Why not the others: (A) reverses the effect of inward sliding. (B) substitutes the constant A-band length for H-zone width. (C) calculates total non-A-band length rather than the central gap between thin-filament tips.

Key point: Identify which filament boundaries define a band before using a length difference.

Q14 / B What separate pH-rate profiles can and cannot establish

Answer: (B) Different optimal protonation environments are supported.

Why: pH affects ionisation of groups needed for active-site structure and catalysis. The separated peaks indicate different conditions for maximal measured activity. Because the substrates differ and the measurements use saturating concentrations, the plot does not provide the concentration dependence required to compare affinity. Nor does a pH profile alone distinguish reversible ionisation changes from irreversible damage after every possible exposure.

Why not the others: (A) infers molecular mass from an activity optimum without a mass measurement. (C) invents an interaction between enzymes assayed separately. (D) turns reduced rate away from an optimum into universal irreversible inactivation.

Key point: A pH-activity profile measures condition-dependent catalysis, not every molecular property of an enzyme.

Q15 / C Shared-species threshold for separated versus contiguous area

Answer: (C) $k < 64 - 32\sqrt{2}$.

Given: Richness doubles when area quadruples; each separate patch has 32 species.

Formula/Principle: Infer the power-law exponent and compare a set union with contiguous-area richness.

Step 1:

$$2 = 4^z \;\Rightarrow\; z = 1/2.$$

Step 2: For one 8A patch,

$$S_{8A} = 16\sqrt{8} = 32\sqrt{2}.$$

Step 3: For the two-patch union, overlapping species are counted once:

$$S_{\text{union}} = 32 + 32 - k = 64 - k.$$

Step 4: Strict superiority requires

$$64 - k > 32\sqrt{2} \;\Rightarrow\; k < 64 - 32\sqrt{2} \approx 18.75.$$

For integer k the allowed values are 0 through 18.

Why: Doubling sampled area does not double richness when the exponent is below one. Separate patches also cannot have their species totals added without accounting for shared species. Under the explicitly assumed common habitat law, the comparison depends jointly on area scaling and overlap, not on area alone; it is not a general claim about which reserve design is always superior.

Why not the others: (A) reverses the superiority inequality. (B) compares overlap with the contiguous richness rather than subtracting it from 64. (D) excludes many positive overlaps that still meet the strict bound.

Key point: Combine species-area scaling with unique-species accounting before comparing spatial samples.

Chemistry

Q16 / C Amide planarity and rotation as evidence of partial C-N multiple bonding

Answer: (C) Delocalisation gives partial C-N double-bond character.

Why: Donation of the nitrogen lone pair into the carbonyl system links the nitrogen and carbonyl orbitals. A planar arrangement supports overlap and gives the actual C-N bond more multiple-bond character than an isolated amine C-N bond. Twisting reduces that overlap, costing energy. The charged drawing is a contributor to one delocalised electronic structure, not evidence that the molecule spends alternating intervals in separately isolable forms.

Why not the others: (A) replaces a resonance hybrid by a kinetic equilibrium of drawings. (B) violates the ordinary nitrogen valence count; the contributor has an octet, not five bonds. (D) contradicts the carbonyl connectivity retained in both contributors.

Key point: Use resonance to explain measurable bond properties without treating contributors as separate molecules.

Q17 / A Global charge balance and local donor counting in a bridged complex

Answer: (A) +3 and coordination number 6.

Given: Four neutral en ligands, two hydroxide ions and overall charge +4; the two metals are equivalent.

Formula/Principle: Charge balance determines oxidation state; count donor atoms attached to each metal for coordination number.

Step 1:

$$2x + 4(0) + 2(-1) = +4 \;\Rightarrow\; x = +3.$$

Step 2: Each cobalt receives four N donors from its two en ligands and two O donors from the bridges:

$$\text{CN} = 4 + 2 = 6.$$

Why: A bridging oxygen counts as one donor to each metal it actually contacts. Ligand count for the entire ion is therefore not the coordination number of either cobalt. Likewise a neutral chelator contributes donor atoms without contributing ionic charge.

Why not the others: (B) neglects the hydroxide charges in oxidation-state balance. (C) counts only chelate donors and omits the bridges. (D) counts the whole ion's chelate donors at one metal and assigns the wrong charge.

Key point: Count charge globally but donor contacts locally in a multinuclear complex.

Q18 / D Allowed excitation versus ground-state occupancy and unpaired spins

Answer: (D) P has two unpaired electrons, Q four, and Q is excited.

Why: P has a paired 2s orbital and two singly occupied p orbitals, giving two unpaired electrons. Q has a singly occupied 2s and three singly occupied p orbitals, giving four. Both preserve six electrons and obey Pauli exclusion, but Q promotes an electron from lower-energy 2s to 2p in this multielectron atom. The promotion costs energy even though it increases the number of unpaired electrons.

Why not the others: (A) confuses an excited state with an exclusion-rule violation. (B) assumes electron count alone fixes electronic energy. (C) reverses the occupations shown.

Key point: Check allowed occupancy separately from whether a configuration is the lowest-energy one.

Q19 / B Oxygen protonation followed by carbonyl-carbon activation

Answer: (B) Oxygen donates to H+ and activates the carbonyl carbon.

Why: A nucleophile supplies the electron pair used to form a new bond. Here the carbonyl oxygen is the donor and the proton is the electron-pair acceptor. Protonation increases the electron deficiency of the conjugated carbonyl system, making carbon more susceptible to a subsequent nucleophilic attack. Roles are assigned for the specific step; an atom's role in one event does not require it to be the attack site in the next.

Why not the others: (A) assigns pair donation to the wrong atom and reverses the activation effect. (C) gives a proton an electron pair it does not possess and introduces radical chemistry. (D) changes the carbon skeleton without any carbon-cleavage step.

Key point: Follow electron-pair origin and destination rather than assigning nucleophile and electrophile labels by charge alone.

Q20 / C Condition-dependent side-chain radical versus ring electrophilic chlorination

Answer: (C) X gives benzyl chloride and Y ring-chlorinated ortho/para products.

Why: Light promotes radical halogenation at the benzylic side chain, where radical stabilisation supports replacement of a methyl hydrogen. Lewis-acid-assisted chlorination instead supplies an electrophile for aromatic substitution. The pre-existing methyl group directs that ring substitution mainly to ortho and para positions. The directing group relevant to the first ring substitution is methyl, not a chlorine atom that has not yet been attached.

Why not the others: (A) reverses the stated radical and electrophilic conditions. (B) ignores the change in reaction mechanism caused by illumination and catalyst. (D) uses the wrong directing group and wrongly makes its deactivation imply meta direction.

Key point: Reaction conditions can select side-chain versus ring chemistry in the same aromatic substrate.

Q21 / A Mixed Wurtz coupling and competing carbon-group pairings

Answer: (A) Ethane, propane and butane can form.

Why: Joining two methyl groups gives ethane, joining methyl with ethyl gives propane, and joining two ethyl groups gives butane. A mixed-halide Wurtz reaction therefore does not uniquely specify the desired cross-coupling product. The argument concerns competing carbon-group pairings, not an inability to make any propane. It also explains why symmetric alkane preparation is more straightforward with a single suitable halide.

Why not the others: (B) invents carbon removal instead of coupling. (C) excludes the cross-pairing that is explicitly allowed. (D) ignores the even-carbon products of both self-pairings.

Key point: List all allowed coupling partners before treating a mixed-halide reaction as a selective synthesis.

Q22 / D Two ether cleavages followed by conversion of an intermediate aliphatic alcohol

Answer: (D) Phenol, 1,2-diiodoethane and methyl iodide; 3 mol HI.

Given: Both ether links are cleaved, ordinary aliphatic OH is converted further, and aryl-O cleavage is excluded.

Formula/Principle: HI cleaves alkyl-O bonds; a generated aliphatic alcohol can consume another HI to form an iodide.

Step 1: Cleaving the phenoxyethyl bond can give

$$\mathrm{Ph{-}O{-}CH_2{-}CH_2{-}O{-}CH_3 + HI \longrightarrow PhOH + I{-}CH_2{-}CH_2{-}O{-}CH_3.}$$

Step 2: Attack at the methyl end gives

$$\mathrm{I{-}CH_2{-}CH_2{-}O{-}CH_3 + HI \longrightarrow I{-}CH_2{-}CH_2{-}OH + CH_3I.}$$

Step 3: Continued reaction converts the remaining alcohol:

$$\mathrm{I{-}CH_2{-}CH_2{-}OH + HI \longrightarrow I{-}CH_2{-}CH_2{-}I + H_2O.}$$

Step 4: Summing consumes three HI and conserves all nine carbons among phenol, the two-carbon diiodide and methyl iodide. A different order of ether cleavages gives the same final set under the stated completion conditions.

Why: The resistant aryl-O bond leaves phenol as the aryl-containing product. Stopping after the two original ether cleavages would overlook the reactive aliphatic OH formed along the route. The explicit assumptions exclude competing elimination so that functional-group and atom accounting identify a single final set.

Why not the others: (A) breaks the excluded aryl-O bond and invents reduction to methane. (B) stops at a reactive alcohol before the stated completion. (C) replaces two C-I substitutions by reduction to ethane.

Key point: Under excess-reagent conditions, continue through reactive intermediates before counting final products and equivalents.

Q23 / C An ester identified by a separated alcohol's diagnostic oxidation test

Answer: (C) Ethyl ethanoate.

Why: Among the proposed C4 esters, ethyl ethanoate releases ethanol, which is oxidised under iodoform conditions to the required methyl-carbonyl system. Methyl propanoate releases methanol, and propyl methanoate releases propan-1-ol, neither of which gives that standard positive test. Separation matters because the test is being assigned specifically to the alcohol fragment rather than to the original mixture. The molecular formula independently excludes the C3 ester.

Why not the others: (A) gives methanol rather than ethanol. (B) gives propan-1-ol, which lacks the required alcohol arrangement for a positive test. (D) has formula C3H6O2 and also gives methanol.

Key point: Use hydrolysis to map an ester's two carbon fragments before applying a diagnostic product test.

Q24 / A Diazonium cyanation as carbon addition followed by nitrogen removal

Answer: (A) Diazotisation, cyanide substitution and nitrile hydrolysis.

Why: Diazotisation converts the amino group into a leaving diazonium group. Copper-cyanide substitution introduces a carbon-containing CN group directly on the ring, giving benzonitrile and adding one carbon to the original six. Hydrolysis converts the nitrile carbon into a carboxyl carbon and removes its nitrogen into an inorganic nitrogen-containing product. The carbon count therefore increases while the final benzoic acid contains no nitrogen.

Why not the others: (B) removes the amino substituent without introducing the extra carbon. (C) misapplies Hofmann degradation and does not place a carboxyl group on the ring. (D) only changes a halogen substituent and supplies no new carbon.

Key point: Cyanide substitution followed by hydrolysis is a carbon-adding route to an aromatic carboxylic acid.

Q25 / B Limits of formula identification from a complete-dissociation particle count

Answer: (B) Five ions form per formula unit, without unique formula determination.

Why: Under the specified complete-dissociation model, i is the ratio of solute-particle amount to analytical formula-unit amount. It therefore counts the total number of separated ions produced. Several charge-balanced stoichiometries can supply five ions, such as A2B3 with suitable 3+/2- charges or AB4 with suitable 4+/1- charges. A colligative measurement by itself does not identify ionic charges or chemical elements.

Why not the others: (A) claims uniqueness without charge or composition evidence. (C) confuses particle number with charge magnitude. (D) equates the number of particles with the number of distinct elements.

Key point: A colligative particle count can constrain stoichiometry without uniquely identifying a salt.

Q26 / C Cell calibration followed by finite-dissociation equilibrium inference

Answer: (C) 8.33×10^{-4} M.

Given: Reference conductivity and resistance calibrate one fixed cell; acid concentration and limiting molar conductivity are supplied.

Formula/Principle: Cell constant $g = \kappa R$, $\Lambda_m = 1000\kappa/c$, $\alpha = \Lambda_m/\Lambda_m^\infty$, and $K_a = c\alpha^2/(1-\alpha)$.

Step 1:

$$g = 0.0140(100) = 1.40 \text{ cm}^{-1}, \quad \kappa_{HA} = 1.40/1400 = 0.00100 \text{ S/cm}.$$

Step 2:

$$\Lambda_m = 1000(0.00100)/0.0100 = 100 \text{ S cm}^2/\text{mol}, \quad \alpha = 100/400 = 0.250.$$

Step 3:

$$K_a = \frac{0.0100(0.250)^2}{0.750} = 8.33 \times 10^{-4} \text{ M}.$$

Why: Resistance depends on cell geometry, so the reference must first establish the cell constant. The molar conductivity ratio estimates the ionised fraction under the ideal model. A 25% dissociation is too large to silently replace 1-alpha by one, so the full denominator is retained.

Why not the others: (A) uses c times alpha rather than the equilibrium concentration ratio. (B) omits the undissociated-acid denominator. (D) assumes complete acid dissociation from the existence of a conductive solution.

Key point: Calibrate conductance geometry before extracting dissociation and retain a non-negligible ionisation correction.

Q27 / B Arrhenius-ratio interpolation combined with unequal reactant concentrations

Answer: (B) 6.

Given: Both reactions are first order; two rate-constant ratios constrain their Arrhenius difference.

Formula/Principle: $\ln(k_X/k_Y) = \ln(A_X/A_Y) - (E_X - E_Y)/(RT)$ is linear in 1/T, not T.

Step 1: The reciprocal temperature at 400 K is the midpoint:

$$\frac{1}{400} = \frac{1}{2}\left(\frac{1}{300} + \frac{1}{600}\right).$$

Step 2: Consequently

$$\ln(k_X/k_Y)_{400} = \tfrac{1}{2}(\ln 4 + \ln 1) = \ln 2,$$

so the rate-constant ratio is 2.

Step 3: Initial rates include concentration:

$$\frac{v_X}{v_Y} = \frac{k_X[X]_0}{k_Y[Y]_0} = 2(3) = 6.$$

Step 4: As a consistency check, the ratio declines on warming, so the X reaction has the smaller activation energy under these assumptions.

Why: Comparing two Arrhenius equations eliminates the need to know either absolute rate constant. Interpolation must use inverse temperature because that is the linear variable. A rate-constant ratio becomes a reaction-rate ratio only after including the concentration factors and the stated first-order law.

Why not the others: (A) ignores the rate-constant difference. (C) reports the rate-constant ratio alone. (D) uses the 300 K ratio at 400 K.

Key point: Interpolate Arrhenius ratios in reciprocal temperature, then distinguish rate constants from rates.

Q28 / D Equilibrium-constant algebra and the resulting standard Gibbs sign

Answer: (D) K=1/2 and the standard Gibbs change is positive.

Given: Two equilibria share A at one temperature.

Formula/Principle: Reaction addition multiplies constants; reversal inverts K and doubling squares it.

Step 1: Reverse and double the first reaction:

$$2\,\mathrm{B} \rightleftharpoons 2\,\mathrm{A}, \quad K' = 1/4^2 = 1/16.$$

Step 2: Add A $\rightleftharpoons$ C, cancelling one A:

$$K = (1/16)(8) = 1/2.$$

Step 3:

$$\Delta G^\circ = -RT \ln K = RT \ln 2 > 0.$$

Why: Equilibrium constants belong to specified stoichiometric equations. Changing coefficients changes the exponent pattern in the activity quotient. The standard Gibbs sign is then determined by whether the resulting dimensionless K lies above or below one; it does not specify the actual direction for every nonstandard composition.

Why not the others: (A) divides by K1 only once instead of squaring it. (B) multiplies the original constants without reversing or doubling. (C) reverses the sign in the Gibbs-constant relation.

Key point: Manipulate the reaction equation before manipulating its equilibrium constant.

Q29 / A Water-dominated charge balance in an extremely dilute weak acid

Answer: (A) Slightly below 7.

Why: The acid concentration is smaller than the hydrogen-ion concentration of pure water. Charge balance requires hydrogen ions to balance both hydroxide and the acid anion, so acid addition cannot make this solution alkaline. Most of this very dilute acid is ionised because Ka greatly exceeds the resulting hydrogen-ion concentration, but water remains a major source of both ions. Neglecting water or assuming a small ionised fraction is therefore inconsistent.

Given: $C = 10^{-8}$ M, $K_a = 10^{-5}$, $K_w = 10^{-14}$.

Formula/Principle: Charge balance is $[H^+] = [OH^-] + [A^-]$, with $0 < [A^-] < C$.

Step 1: The bounds imply

$$10^{-7} < [H^+] < \frac{C + \sqrt{C^2 + 4K_w}}{2} \approx 1.052 \times 10^{-7}\ \mathrm{M}.$$

Step 2: Hence $6.978 <$ pH < 7, confirming a small acidic shift.

Why not the others: (B) ignores water and even predicts alkalinity from an acid. (C) uses the small-dissociation approximation where its predicted dissociation exceeds the total acid. (D) assumes a buffer equality that is not present.

Key point: Very dilute acid solutions require water and charge balance even when the acid is weak.

Q30 / D Lanthanoid contraction explains a cross-period chemical resemblance without identity

Answer: (D) Lanthanoid contraction helps explain similar size and chemistry.

Why: Poor shielding by added 4f electrons across the intervening lanthanoid series increases effective nuclear attraction and contracts later atomic and ionic sizes. Hf therefore resembles Zr in size and in many +4 chemical behaviours more closely than a simple extra-shell expectation would suggest. Atomic mass is a separate property and can differ greatly even when ionic radius and common oxidation state are similar.

Why not the others: (A) treats mass as a direct valence rule. (C) contradicts the role of the intervening filled 4f series and confuses Hf chemistry with a claim about preceding atoms. (B) generalises a size/chemistry similarity to every mass-dependent or spectroscopic property.

Key point: A periodic size explanation supports related chemistry without making two elements physically identical.

Mathematics

Q31 / A Ellipse focal distances determine a triangle height rather than a directrix

Answer: (A) $8\sqrt{2}$.

Given: The focal separation is 6 and the sum of focal distances is 10.

Formula/Principle: Use the ellipse's distance sum and subtract squared distance equations to recover the height above the focal axis.

Step 1: The right-focus distance is $10 - 4 = 6$. Write

$$(x+3)^2 + y^2 = 16, \qquad (x-3)^2 + y^2 = 36.$$

Step 2: Subtraction gives $12x = -20$, so $x = -5/3$. Then

$$y^2 = 16 - (4/3)^2 = 128/9, \quad y = 8\sqrt{2}/3$$

because P is on the upper half.

Step 3:

$$\mathcal{A} = \tfrac{1}{2}(6)(8\sqrt{2}/3) = 8\sqrt{2}.$$

Worked derivation:

$$\mathcal{A} = 3y = 8\sqrt{2}, \qquad y > 0.$$

Why: The ellipse supplies the missing second distance, after which the difference of squares eliminates the unknown ordinate. The focal axis is a natural triangle base, but its height is y, not either focal distance. The upper-half condition removes the sign ambiguity without changing the area.

Why not the others: (B) halves the correct triangle area again. (C) treats the major-axis length as an area. (D) omits the triangle's one-half factor.

Key point: Convert focal-distance data into coordinates before using a geometric height.

Q32 / D Flat zero crossing inside an absolute value preserves two derivatives

Answer: (D) Twice differentiable with second derivative zero.

Given: $x - \sin x$ has the sign of x for nonzero x near zero.

Formula/Principle: Resolve the absolute value and evaluate the derivative definitions at the join.

Step 1: For x positive, $f = x - \sin x$; for x negative, $f = \sin x - x$. Also

$$\frac{f(x)}{x} \to 0$$

because $\sin x/x \to 1$, so f'(0)=0.

Step 2: Away from zero,

$$f'(x) = \begin{cases} 1 - \cos x, & x > 0, \\ \cos x - 1, & x < 0. \end{cases}$$

Step 3: Using $(1 - \cos x)/x \to 0$, both sides of $[f'(x) - f'(0)]/x$ tend to zero. Hence f"(0)=0.

Worked derivation:

$$f''(0) = \lim_{x \to 0} \frac{f'(x)}{x} = 0.$$

Why: An absolute value creates a corner only when its inner expression crosses zero with a sufficiently nonzero first-order slope. Here the inner expression has a flat enough contact at zero for two derivatives to survive. Its sign follows from the nonnegative derivative 1-cos x and its zero value at the origin; no power-series expansion is required.

Why not the others: (A) assumes every absolute value has a corner at an inner zero. (B) stops after the first-derivative test despite the second derivative limit existing. (C) incorrectly transfers the second derivative of another elementary function to this joined expression.

Key point: Test derivatives at the join instead of deciding smoothness from the presence of an absolute value.

Q33 / B A parameter-dependent integration breakpoint and exhaustive real-solution check

Answer: (B) -1/2.

Given: The absolute-value break depends on the unknown a.

Formula/Principle: Separate parameters placing the break inside the interval from those placing it outside.

Step 1: For $-1 \le a \le 1$,

$$I(a) = \int_{-1}^{a} (a - x)\,dx + \int_{a}^{1} (x - a)\,dx = 1 + a^2.$$

Step 2: Setting this equal to 3/2 gives $a = \pm 1/\sqrt{2}$, both within the assumed interval.

Step 3: For $|a| \ge 1$, $I(a) = 2|a| \ge 2$, so there are no additional roots.

Worked derivation:

$$\prod a = \left(\frac{1}{\sqrt{2}}\right)\left(-\frac{1}{\sqrt{2}}\right) = -\frac{1}{2}.$$

Why: The integration formula changes when the moving zero leaves the integration interval. Checking the outer parameter ranges proves the two inner solutions exhaust the real line. The requested product is negative because the two valid parameters have opposite signs.

Why not the others: (A) uses the outer-range formula where its proposed parameters violate that case. (C) discards the sign of the negative solution. (D) confuses the zero sum with the product.

Key point: When an integration breakpoint is a parameter, validate every parameter region.

Q34 / C Counting right inverses by finite preimage fibres

Answer: (C) 4.

Given: Each value of g must be chosen from the corresponding preimage under f.

Formula/Principle: Independent choices of one preimage in each nonempty fibre count right inverses.

Step 1:

$$g(0) = 0, \quad g(1) \in \{-1, 1\}, \quad g(4) \in \{-2, 2\}.$$

Step 2:

$$N = 1 \cdot 2 \cdot 2 = 4.$$

Worked derivation:

$$|f^{-1}(\{0\})|\,|f^{-1}(\{1\})|\,|f^{-1}(\{4\})| = 1(2)(2) = 4.$$

Why: f is onto but not one-to-one. A right inverse can still choose one representative from each preimage set even though a two-sided inverse for f is impossible on the full domain. The equation specified in the question is the right-inverse condition only.

Why not the others: (A) counts the sign choice for only one nonzero target. (B) wrongly gives zero two preimages as well. (D) requires f to be injective for a condition that only requires surjectivity.

Key point: Distinguish a chosen right inverse from a two-sided inverse.

Q35 / A An adjugate constraint determines a block matrix and two trace values

Answer: (A) -2 and 6.

Given: The matrix form and adjugate identity hold entry by entry.

Formula/Principle: Compute cofactors without assuming invertibility in advance.

Step 1:

$$\operatorname{adj} A = \begin{pmatrix} 2a & -2b & 0 \\ -2b & 2a & 0 \\ 0 & 0 & a^2 - b^2 \end{pmatrix}.$$

Step 2: Off-diagonal equality $-2b = 2b$ gives b=0; the bottom-right equality gives $a^2 - b^2 = 4$, so $a = \pm 2$.

Step 3:

$$\operatorname{tr} A = 2a + 2 \in \{-2, 6\}.$$

Worked derivation:

$$A = \operatorname{diag}(2, 2, 2) \text{ or } \operatorname{diag}(-2, -2, 2), \quad \operatorname{adj} A = 2A.$$

Why: A matrix identity supplies several coupled scalar conditions. The off-diagonal sign in the adjugate is essential and eliminates b before the quadratic in a is solved. Both signs of a satisfy every entry condition, so assuming positive diagonal entries would lose a genuine matrix.

Why not the others: (B) miscalculates the trace when a=-2. (C) changes the fixed last diagonal entry to -2. (D) discards the negative a solution without a stated restriction.

Key point: Solve adjugate identities entrywise and verify every algebraic branch.

Q36 / B Independent events conditioned on their exclusive union

Answer: (B) 2/5.

Given: Independence applies before the exactly-one condition is imposed.

Formula/Principle: The conditioning event is the disjoint union of A-only and B-only.

Step 1:

$$P(A \text{ only}) = 0.4(1-0.5) = 0.2, \quad P(B \text{ only}) = 0.5(1-0.4) = 0.3.$$

Step 2:

$$P(A \mid \text{exactly one}) = \frac{0.2}{0.2+0.3} = \frac{2}{5}.$$

Worked derivation:

$$P = \frac{P(A \cap B^c)}{P(A \cap B^c) + P(A^c \cap B)} = \frac{0.2}{0.5}.$$

Why: Exactly one excludes both the intersection and the neither event. Original independence permits the two product calculations, but does not make the two surviving alternatives equiprobable. Conditional probability must use their actual masses.

Why not the others: (A) assumes two named alternatives have equal probability. (C) reports the unnormalised A-only probability. (D) divides P(A), including the forbidden intersection, by the exactly-one probability.

Key point: Apply independence to compute outcomes, then renormalise the retained event.

Q37 / D Gap-constrained lattice paths with a forbidden vertex

Answer: (D) 3.

Given: Four R steps define five gaps in which nonconsecutive U steps may be placed.

Formula/Principle: Choose three distinct gaps, then apply the forbidden-vertex condition to those gap choices.

Step 1: Label gaps 0 through 4: before the first R, between consecutive R steps, and after the last R. Three selected gaps give all $\binom{5}{3} = 10$ paths with no consecutive U.

Step 2: If exactly one of gaps 0 and 1 is occupied, the path is at (2,1) immediately after its second R, so it is forbidden.

Step 3: If neither gap is occupied, all of gaps 2,3,4 must be occupied. The U in gap 2 then passes through (2,1), also forbidden.

Step 4: The only avoiding paths occupy both gaps 0 and 1, with the third U in any of gaps 2,3,4. Hence there are 3.

Worked derivation:

$$N = \binom{3}{1} = 3, \quad \text{selected gaps } \{0,1,2\}, \{0,1,3\}, \{0,1,4\}.$$

Why: Ordinary unrestricted lattice-path subtraction would include paths with consecutive up steps, so the two constraints must be handled in the same representation. The gap description makes adjacency automatic and locates the forbidden vertex precisely, including a vertical step through it.

Why not the others: (A) ignores the forbidden vertex. (B) counts the rejected gap selections. (C) treats an unavailable extra gap as a possible third U position.

Key point: Choose a representation in which an adjacency rule is automatic before enforcing a geometric exclusion.

Q38 / A A weighted binomial identity inferred by finite integration before locating a middle term

Answer: (A) 70.

Given: The weighted alternating sum runs over the full binomial expansion.

Formula/Principle: Integrate the finite expansion of $(1-x)^n$ from zero to one.

Step 1:

$$\int_0^1 (1-x)^n\,dx = \sum_{k=0}^{n}(-1)^k\binom{n}{k}\int_0^1 x^k\,dx = \sum_{k=0}^{n}\frac{(-1)^k}{k+1}\binom{n}{k}.$$

Step 2: Direct integration gives

$$\frac{1}{n+1} = \frac{1}{9} \Rightarrow n = 8.$$

Step 3: There are nine terms, so the middle coefficient is

$$\binom{8}{4} = 70.$$

Worked derivation:

$$\binom{8}{4} = \frac{8\cdot 7\cdot 6\cdot 5}{4\cdot 3\cdot 2\cdot 1} = 70.$$

Why: The reciprocal k+1 weight points to integration of x to the kth power, whereas ordinary substitution into the binomial expansion would not create that denominator. The sum is finite, so term-by-term integration needs no infinite-series convergence argument.

Why not the others: (B) uses a neighbouring coefficient of the wrong row. (C) uses the middle coefficient of row 9 after confusing n with n+1. (D) selects coefficient k=2 of row 8 instead of its middle.

Key point: Reciprocal index weights in finite binomial sums can be generated by an elementary integral.

Q39 / C A self-referential partial-sum constraint becomes a shifted geometric progression

Answer: (C) $3(3/2)^{n-1} - 2$.

Given: The partial-sum relation holds at every positive integer, including n=1.

Formula/Principle: Difference consecutive partial sums, then shift the resulting affine recurrence into a GP.

Step 1: At n=1,

$$a_1 = 3a_1 - 2 \Rightarrow a_1 = 1.$$

Step 2: For n at least 2,

$$a_n = S_n - S_{n-1} = 3a_n - 3a_{n-1} - 2,$$

hence $a_n = \frac{3}{2}a_{n-1} + 1$.

Step 3: Put $b_n = a_n + 2$. Then

$$b_n = \tfrac{3}{2}b_{n-1}, \qquad b_1 = 3.$$

Step 4: Thus

$$b_n = 3(3/2)^{n-1}, \qquad a_n = 3(3/2)^{n-1} - 2.$$

Worked derivation:

$$\sum_{j=1}^{n} a_j = 6[(3/2)^n - 1] - 2n = 3a_n - 2n,$$

directly verifying the full partial-sum identity.

Why: Differencing does not simply remove the n-dependent term because a_n appears both inside the sum and on its right-hand side. The constant shift finds the fixed offset of the affine recurrence, leaving an ordinary geometric progression. The n=1 equation supplies the indispensable initial value.

Why not the others: (A) uses the wrong sign of the offset and fails at n=1. (B) drops the additive term in the recurrence. (D) inverts the derived common ratio.

Key point: A partial-sum relation can generate an affine recurrence whose shifted terms form a GP.

Q40 / D Continuity at one point plus a scaling identity establishes the derivative

Answer: (D) 2.

Given: Only continuity at zero is assumed initially, not differentiability.

Formula/Principle: Halve the argument repeatedly and sum a finite telescoping identity before taking a limit.

Step 1: Replacing x by x/2 gives

$$f(x) - f(x/2) = x.$$

Step 2: Summing the halved versions yields

$$f(x) - f(x/2^n) = x \sum_{j=0}^{n-1} 2^{-j} = 2x(1 - 2^{-n}).$$

Step 3: Continuity at zero makes $f(x/2^n) \to 1$, so

$$f(x) = 1 + 2x.$$

Step 4: The derivative at zero therefore exists and equals 2.

Worked derivation:

$$f'(0) = \lim_{h \to 0} \frac{(1 + 2h) - 1}{h} = 2.$$

Why: Differentiating the original identity immediately would assume the very regularity that must be established. Repeated halving carries each fixed real x to the one point where continuity is known, and a finite geometric sum recovers the whole function. The argument excludes extra oscillatory solutions that would violate continuity at zero.

Why not the others: (A) confuses continuity with a horizontal slope. (B) uses the value f(0) as the derivative. (C) introduces an extra factor from the doubled argument without applying the chain relation consistently.

Key point: Use known continuity to establish a function before assuming its differentiability.

Q41 / B Homogeneous substitution followed by a domain-checked first zero

Answer: (B) e.

Given: The differential equation depends on the ratio y/x and x is positive.

Formula/Principle: Use the homogeneous substitution $v = y/x$, so $y' = v + xv'$.

Step 1:

$$x(v + xv') = xv - x \Rightarrow v' = -1/x.$$

Step 2: Integration and y(1)=1 give

$$v = 1 - \ln x, \quad y = x(1 - \ln x).$$

Step 3: For x>0 the zero requires $\ln x = 1$, giving the unique positive zero e.

Worked derivation:

$$y(e) = e(1 - 1) = 0, \qquad xy' = x(-\ln x) = y - x.$$

Why: The substitution isolates the changing ratio rather than treating y/x as a constant. The positive domain justifies the logarithm and excludes x=0 as a spurious zero of the formula. The explicit check confirms both the equation and the initial value.

Why not the others: (A) is the initial point where y=1. (C) uses the wrong logarithmic sign. (D) doubles the exponent without a second condition.

Key point: Preserve the original positive domain when finding zeros of a homogeneous-equation solution.

Q42 / D A spatial half-turn recovered through the foot on its axis

Answer: (D) 19/9.

Given: The line has base point A=(1,0,1) and direction d=(1,2,2).

Formula/Principle: Find the perpendicular foot H on the line; a half-turn sends P to $P' = 2H - P$.

Step 1:

$$t_H = \frac{(P-A)\cdot \mathbf{d}}{|\mathbf{d}|^2} = \frac{(1,1,-1)\cdot(1,2,2)}{9} = 1/9.$$

Step 2:

$$H = (10/9, 2/9, 11/9).$$

Step 3:

$$P' = 2H - P = (2/9, -5/9, 22/9).$$

Step 4: The coordinate sum is $(2-5+22)/9 = 19/9$.

Worked derivation:

$$(P-H)\cdot \mathbf{d} = 0, \qquad (P+P')/2 = H, \qquad \sum P'_i = 19/9.$$

Why: A reflection across a spatial line here is explicitly a half-turn, not reflection in a plane containing the line. The parallel component stays fixed while the entire perpendicular component reverses, making H the midpoint. Both orthogonality and midpoint checks are required to verify the construction.

Why not the others: (A) adds incorrect reflected components. (B) reports the line parameter rather than a coordinate sum. (C) retains the original point's coordinate sum without applying the half-turn.

Key point: Define a spatial line reflection precisely and use its perpendicular midpoint.

Q43 / C A cross-product equality forces coplanarity including degenerate cases

Answer: (C) Zero.

Given: The same a has equal cross products with b and c; zero vectors and equal b,c are allowed.

Formula/Principle: Zero cross product means parallel vectors or a zero vector, and volume is the absolute scalar triple product.

Step 1:

$$\mathbf{a} \times (\mathbf{b} - \mathbf{c}) = 0.$$

Step 2: If a=0 or b=c, the triple product is immediately zero. Otherwise a is parallel to b-c and hence lies in the span of b,c.

Step 3:

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = 0.$$

Worked derivation:

$$V = |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})| = 0.$$

Why: Equal cross products constrain the relative geometry, not merely two areas. After subtraction, the parallelism condition forces coplanarity, including all degenerate cases. No assumption of three independent directions is compatible with the supplied equality.

Why not the others: (A) omits the angular factor between a and b cross c and need not vanish. (B) invents a magnitude without any scale data. (D) overlooks the forced coplanarity.

Key point: A vector-product identity can rule out nonzero volume before any coordinates are introduced.

Q44 / A An unknown frequency changes the mean-deviation sign regions

Answer: (A) 6.

Given: Total frequency is $N = p + 6$ and weighted sum is 20.

Formula/Principle: Mean deviation uses absolute distances from the mean, whose position depends on p.

Step 1:

$$\mu = 20/(p+6), \qquad 0 < \mu \le 10/3.$$

The mean lies at or below 2 exactly when p is at least 4.

Step 2: For p at least 4,

$$D = \frac{p\mu + 4(2-\mu) + 2(6-\mu)}{p+6} = \frac{40p}{(p+6)^2}.$$

Equating to 5/3 gives

$$(p+6)^2 = 24p \Rightarrow (p-6)^2 = 0,$$

hence p=6.

Step 3: For integer p=0,1,2,3, the mean is above 2 and

$$D = \frac{p\mu + 4(\mu-2) + 2(6-\mu)}{p+6} = \frac{24p+64}{(p+6)^2}.$$

The same target gives $5p^2 - 12p - 12 = 0$, with roots $(6 \pm 4\sqrt{6})/5$, neither an integer.

Step 4: At p=6, $\mu = 5/3$, and the direct absolute-deviation sum is 20 over 12 observations, giving 5/3.

Worked derivation:

$$D = \frac{6(5/3) + 4(1/3) + 2(13/3)}{12} = \frac{5}{3}.$$

Why: The distances for value 2 reverse sign when the mean passes through 2. Solving only one unsigned expression can introduce or miss frequency solutions. The integer restriction rejects the second case's real root; checking p=4 in the first case covers the boundary without a gap.

Why not the others: (B) lies in the upper-mean case but fails its equation. (C) places the mean at 2 and gives deviation 8/5. (D) gives a different mean and deviation despite also satisfying the nonnegative-frequency requirement.

Key point: When frequencies determine the mean, locate that mean before removing absolute values.

Q45 / B Symmetric tangent constraints determine a cosine difference without solving angles

Answer: (B) $3/\sqrt{37}$.

Given: Both cosines are positive because the angles are acute.

Formula/Principle: Express the cosine difference using tangents and obtain the denominator from symmetric sums.

Step 1: Put u=tan alpha and v=tan beta. Then

$$\cos(\alpha - \beta) = \frac{1+uv}{\sqrt{(1+u^2)(1+v^2)}}.$$

Step 2:

$$u^2 + v^2 = (u+v)^2 - 2uv = 9 - 1 = 8,$$

so

$$(1+u^2)(1+v^2) = 1 + 8 + 1/4 = 37/4.$$

Step 3:

$$\cos(\alpha-\beta)=\frac{3/2}{\sqrt{37}/2}=\frac{3}{\sqrt{37}}.$$

Worked derivation:

$$u,v=\frac{3\pm\sqrt{7}}{2}>0,\qquad 0<\frac{3}{\sqrt{37}}<1,$$

confirming compatibility with acute angles.

Why: The requested expression is symmetric in the two angles, so solving them individually is unnecessary. Acute-angle information fixes the positive square-root sign when writing cosines in terms of tangents. The symmetric sum and product determine the remaining denominator exactly.

Why not the others: (A) uses 1-uv in the numerator. (C) omits the constant and product-squared terms from the denominator and exceeds one. (D) drops the uv contribution in the numerator.

Key point: Use symmetric tangent data directly and retain the quadrant sign in a cosine formula.

Physics

Q46 / B A cubic platform trajectory determines the return event and impact speed

Answer: (B) -21 m/s.

Given: Ball and platform begin at the same position, with platform initial speed zero.

Formula/Principle: Integrate the platform acceleration and compare positions before subtracting velocities at the nonzero meeting time.

Step 1:

$$v_p=t^2,\qquad y_p=t^3/3.$$

For the ball,

$$v_b=18-10t,\qquad y_b=18t-5t^2.$$

Step 2: Equal positions require

$$18t-5t^2-t^3/3=0\Rightarrow t(t^2+15t-54)=0.$$

Step 3: The quadratic roots are 3 and -18 s; the first positive return is t=3 s, excluding the launch t=0.

Step 4:

$$v_{b/p}=v_b-v_p=(18-30)-9=-21\ \text{m/s}.$$

Worked derivation:

$$y_b(3)=54-45=9\ \text{m}=y_p(3),\quad v_{b/p}(3)=-21\ \text{m/s}.$$

Why: A turning point of the ball or a zero relative velocity is not a meeting event. The platform's acceleration varies, so its displacement is cubic and cannot be obtained by inserting the final acceleration into a constant-acceleration formula. The return velocity must be measured relative to the moving platform, not the ground.

Why not the others: (A) gives the ball's ground-frame velocity only. (C) reports the platform speed with a negative sign. (D) reverses the defined relative-velocity direction.

Key point: Solve a relative-position event before evaluating the relative velocity.

Q47 / C An upper turning point occurs before spring separation

Answer: (C) It rises mg/k and remains in contact.

Given: Initial compression is $x_0 = 3mg/(2k)$; upward displacement z reduces compression to x0-z while contact holds.

Formula/Principle: Conserve mechanical energy, then check whether the turning point precedes the zero-compression point.

Step 1: At a stop during contact,

$$\tfrac{1}{2}kx_0^2 = mgz + \tfrac{1}{2}k(x_0 - z)^2.$$

Step 2: Excluding the release root z=0 gives

$$z = 2(x_0 - mg/k) = mg/k.$$

Step 3: Remaining compression is

$$x_0 - z = mg/(2k) > 0,$$

so the assumed contact is self-consistent and the block has not reached the natural spring length.

Worked derivation:

$$N = k(x_0 - z) = mg/2 > 0 \quad \text{at the first upper turning point.}$$

Why: Launch into free flight requires the block to reach zero spring compression with an allowed kinetic energy. Here it turns earlier, so assigning all initial elastic energy to gravitational height after separation is invalid. The nonzero spring force at the turning point does not imply upward acceleration; it is smaller than weight and permits reversal.

Why not the others: (A) assumes the block necessarily reaches natural length. (B) treats all elastic energy as gravitational energy while ignoring residual spring energy. (D) gives the remaining compression instead of the rise.

Key point: Check the contact boundary before continuing a spring-energy solution into free flight.

Q48 / D A measured rolling-energy fraction identifies dimensionless central inertia

Answer: (D) $2MR^2/3$.

Given: Pure rolling gives $\omega = v/R$, and the translational fraction is 3/5.

Formula/Principle: Add translational and rotational kinetic energy before forming the measured fraction.

Step 1:

$$K = \tfrac{1}{2}Mv^2 + \tfrac{1}{2}I(v/R)^2.$$

Step 2:

$$\frac{K_{\text{trans}}}{K} = \frac{1}{1 + I/(MR^2)} = \frac{3}{5}.$$

Step 3:

$$I/(MR^2) = 5/3 - 1 = 2/3.$$

Worked derivation:

$$K_{\text{rot}}/K_{\text{trans}} = 2/3, \qquad I = 2MR^2/3.$$

Why: Rolling links angular and linear speeds, allowing an energy fraction to reveal mass distribution without naming the body's shape. The inertia is about the centre, because the translational energy has already been included separately. Using an axis at the contact point would double-count that part of the energy.

Why not the others: (A) inverts the rotational-to-translational ratio. (B) equates the translational fraction with the dimensionless inertia. (C) equates the rotational fraction of total energy with the inertia ratio.

Key point: Distinguish rotational energy relative to translational energy from its fraction of the total.

Q49 / A Loss of upper-layer buoyancy makes a block sink further into a denser reservoir

Answer: (A) Downward by H/4.

Given: The initial buoyant contributions each involve half the block volume; the interface height becomes the fixed lower free surface.

Formula/Principle: Infer block density from initial buoyancy and then its new submerged fraction.

Step 1: Initially,

$$\rho_b V g = \rho(V/2)g + (\rho/2)(V/2)g \Rightarrow \rho_b = 3\rho/4.$$

Step 2: In the lower liquid alone,

$$V_{\text{sub}}/V = \rho_b/\rho = 3/4.$$

Step 3: The bottom changes from H/2 below the reference interface to 3H/4 below that same level. The entire block moves down by H/4.

Worked derivation:

$$\Delta z_{\text{bottom}} = -3H/4 - (-H/2) = -H/4.$$

Why: Removing the upper liquid removes a real buoyant contribution, even though the remaining liquid is denser. The block must displace more lower liquid to replace that lost support. Keeping the lower surface fixed is essential to interpreting the change as an absolute vertical displacement.

Why not the others: (B) assumes denser liquid alone necessarily lifts the block. (C) ignores the remaining support and overestimates the shift. (D) overlooks the lost upper-layer buoyancy.

Key point: Compare displaced-fluid contributions before inferring how an interface change moves a floating object.

Q50 / C Irreversible adiabatic expansion against a fixed load requires endpoint energy accounting

Answer: (C) 11V/5.

Given: The expansion is adiabatic but irreversible; external pressure is constant.

Formula/Principle: Use $\Delta U = -P_{\text{ext}}(V_f - V)$ and monatomic ideal-gas $U = 3pV/2$.

Step 1: Endpoint internal energies are

$$U_i = \tfrac{3}{2}(3PV), \qquad U_f = \tfrac{3}{2}(PV_f).$$

Step 2: First-law balance gives

$$\tfrac{3}{2}PV_f - \tfrac{9}{2}PV = -P(V_f - V).$$

Step 3:

$$\tfrac{5}{2}V_f = \tfrac{11}{2}V \Rightarrow V_f = 11V/5.$$

Worked derivation:

$$W = P(V_f - V) = 6PV/5, \quad \Delta U = -6PV/5, \quad Q = 0.$$

Why: The gas is not in mechanical equilibrium with the external pressure throughout the sudden expansion. Work must therefore use the imposed external load, while internal energy uses equilibrium endpoint states. The reversible relation pV to the gamma is not valid for this path, even though no heat is transferred.

Why not the others: (A) imposes isothermal endpoint temperatures. (B) uses the reversible adiabatic formula for an irreversible path. (D) does not satisfy the first-law endpoint balance.

Key point: Adiabatic does not imply reversible; choose the work law from the actual external constraint.

Q51 / B A wave snapshot plus local transverse velocity fixes propagation direction and speed

Answer: (B) Leftward at 10 m/s.

Given: At x=0, the displacement slope is positive and the particle's vertical velocity is positive.

Formula/Principle: For a rightward wave $y_t = -cy_x$; for a leftward wave $y_t = +cy_x$, with c positive.

Step 1:

$$k = 2\pi/0.40 = 5\pi \text{ m}^{-1}, \quad A = 0.020 \text{ m}.$$

Step 2: At the rising zero crossing,

$$y_x = Ak = 0.020(5\pi) = 0.10\pi.$$

Step 3: The signs of yt and yx match, selecting leftward propagation, and

$$c = y_t/y_x = \pi/(0.10\pi) = 10 \text{ m/s}.$$

Worked derivation:

$$y = 0.020 \sin(5\pi x + 50\pi t), \quad y_t(0,0) = \pi \text{ m/s}, \quad c = 50\pi/(5\pi) = 10 \text{ m/s}.$$

Why: A snapshot fixes wavelength and slope but cannot determine propagation direction without a time-sensitive observation. The local particle velocity supplies that missing sign. Wave speed along the string is different from the string element's transverse speed and from the frequency, which here is 25 Hz.

Why not the others: (A) uses the wrong sign relation for a rightward wave. (C) reports the numerical frequency as a speed. (D) combines both the sign and frequency-speed errors.

Key point: Combine a spatial slope with particle motion to infer a travelling wave's direction.

Q52 / D Mutual circular motion separates barycentric radii from gravitational separation

Answer: (D) $8\pi^2 d^3/(3GT^2)$.

Given: Both masses have angular speed $\omega = 2\pi/T$, and the gravitational separation is d.

Formula/Principle: Combine the centre-of-mass radius relation with Newtonian centripetal balance.

Step 1: If m1 is in the smaller orbit,

$$m_1(d/3) = m_2(2d/3) \Rightarrow m_1 = 2m_2.$$

Step 2: For m1,

$$\frac{Gm_1m_2}{d^2} = m_1\omega^2(d/3) \Rightarrow m_2 = \frac{\omega^2 d^3}{3G}.$$

Step 3:

$$m_1 = 2m_2 = \frac{8\pi^2 d^3}{3GT^2}.$$

Worked derivation:

$$m_1 + m_2 = \frac{4\pi^2 d^3}{GT^2}, \qquad m_1 = \frac{2}{3}(m_1 + m_2).$$

Why: The centre of mass is not fixed at either object. The larger mass travels in the smaller circle, while gravitational force depends on the full separation. Using one orbital radius as the force-law separation gives a different and incorrect mass.

Why not the others: (A) gives the smaller mass. (B) gives the total system mass. (C) omits the factor one-third in the centripetal radius.

Key point: Separate orbital radius about the centre of mass from interbody distance in a two-body orbit.

Q53 / C Exact force, torque and energy of a finite dipole in a nonuniform field

Answer: (C) $-3kQq/(16a^2)$, zero torque and $kQq/(4a)$.

Given: The dipole has finite length 2a; the external source is not infinitely far away.

Formula/Principle: Sum the two exact Coulomb forces and charge-potential energies rather than assuming a uniform field.

Step 1: +q is 2a from +Q and is repelled left:

$$F_+ = -kQq/(4a^2).$$

Step 2: -q is 4a from +Q and is attracted right:

$$F_- = +kQq/(16a^2).$$

Step 3:

$$F_x = F_+ + F_- = -3kQq/(16a^2).$$

Both force lines lie on the dipole axis, giving zero torque about the centre.

Step 4:

$$U = (+q)\frac{kQ}{2a} + (-q)\frac{kQ}{4a} = \frac{kQq}{4a}.$$

Worked derivation:

$$F_x = -\frac{3kQq}{16a^2}, \quad \tau = 0, \quad U = \frac{kQq}{4a}.$$

Why: A neutral dipole can experience a net force in a nonuniform field because its two opposite forces have different magnitudes. Zero torque here follows from collinearity, not from zero force. The point-dipole approximation is not exact at these distances, so the finite-charge energies are kept.

Why not the others: (A) reverses the net direction despite the nearer repulsion being stronger. (B) substitutes centre-field approximations for the stated finite geometry. (D) assumes neutrality cancels forces and invents a noncollinear torque.

Key point: For a finite dipole near a source, compute the two charge contributions before invoking a uniform-field approximation.

Q54 / A Dielectric geometry and a capacitance ratio determine material constant and fixed-charge energies

Answer: (A) K=3 and energy ratio 3/4.

Given: Area splitting is parallel combination; thickness splitting is series combination.

Formula/Principle: Add the correct partial capacitances and use $U = Q^2/(2C)$ for equal fixed charges.

Step 1: With $C_0 = \varepsilon_0 A/d$,

$$C_X = C_0(K+1)/2, \qquad C_Y = 2KC_0/(K+1).$$

Step 2:

$$\frac{(K+1)^2}{4K} = \frac{4}{3} \Rightarrow 3K^2 - 10K + 3 = 0 \Rightarrow K = 3 \text{ or } 1/3.$$

Step 3: K>1 retains only K=3. Equal charges give

$$U_X/U_Y = C_Y/C_X = 3/4.$$

Worked derivation:

$$C_X = 2C_0, \quad C_Y = 3C_0/2, \quad U_X/U_Y = (3/2)/2 = 3/4.$$

Why: The two dielectric arrangements share material volume but impose different electric-field paths. Side-by-side portions share voltage; stacked portions share the normal displacement and

behave as series layers in this ideal planar model. Equal-charge energy is inversely proportional to capacitance, reversing the capacitance ratio.

Why not the others: (B) uses the equal-voltage energy ratio. (C) retains the mathematical root excluded by K>1. (D) does not satisfy the measured capacitance ratio.

Key point: Identify area-parallel versus thickness-series geometry and the electrical constraint before comparing energy.

Q55 / D Opposite-corner cube symmetry determines network resistance and a middle-edge current

Answer: (D) Equivalent resistance 5R/6 and marked-edge current V/(5R).

Given: All edges are identical and the terminals are opposite cube vertices.

Formula/Principle: Symmetry makes the three neighbours of each terminal equipotential; reduce the network into three series groups.

Step 1: The three edges out of A are parallel, then six middle edges connect the two equipotential groups, then three edges enter B. Thus

$$R_{\text{eq}} = R/3 + R/6 + R/3 = 5R/6.$$

Step 2: Total current is

$$I = V/R_{\text{eq}} = 6V/(5R).$$

Step 3: It divides equally among the six middle edges:

$$I_{\text{marked}} = I/6 = V/(5R).$$

Worked derivation:

$$V_{\text{middle}} = I(R/6) = V/5, \quad I_{\text{marked}} = (V/5)/R = V/(5R).$$

Why: Equal potentials follow from the terminal-preserving rotational symmetry, not from all cube vertices being identical under the powered circuit. There are six middle edges rather than three, which changes both the equivalent resistance and the individual edge current.

Why not the others: (A) treats all stages as an unrelated single resistance. (B) uses incorrect parallel grouping. (C) finds the equivalent resistance but divides the source voltage across a middle edge as though all six edges were in series.

Key point: Use powered-network symmetry to identify equipotential node groups before reducing a spatial resistor graph.

Q56 / B A reversed third current changes a triangular wire-force sum

Answer: (B) $\mu_0 I^2/(2\pi d)$, parallel to C-to-B.

Given: All pair separations equal d; like currents attract and opposite currents repel.

Formula/Principle: Add the two force-per-length vectors, each of magnitude $f_0 = \mu_0 I^2/(2\pi d)$.

Step 1: B attracts A to the right:

$$\mathbf{f}_{AB} = f_0(1, 0).$$

Step 2: C repels A away from C:

$$\mathbf{f}_{AC} = f_0(-1/2, -\sqrt{3}/2).$$

Step 3:

$$\mathbf{f}_A = f_0(1/2, -\sqrt{3}/2), \quad |\mathbf{f}_A| = f_0.$$

Worked derivation:

$$\mathbf{r}_B - \mathbf{r}_C = d(1/2, -\sqrt{3}/2),$$

proving the force direction is parallel to C towards B.

Why: Reversing one current changes attraction to repulsion without changing that pair's force magnitude. The resulting vector sum is not the inward median appropriate to three equally directed currents. The geometric force directions must be assigned before using a triangle magnitude rule.

Why not the others: (A) gives the result for attraction towards both other vertices. (C) assumes equal force magnitudes imply opposite directions. (D) adds magnitudes while ignoring their different directions.

Key point: Determine current-dependent attraction or repulsion before summing parallel-wire forces geometrically.

Q57 / D A reactive secondary load reflected through an ideal transformer

Answer: (D) 4/3 A and power factor 0.6.

Given: The transformer is ideal and the reactive load has nonzero net inductive reactance.

Formula/Principle: Transform voltage and current magnitudes, while preserving real and apparent power for an ideal transformer.

Step 1:

$$V_s = 120/3 = 40 \text{ V rms.}$$

Step 2:

$$Z = \sqrt{6^2 + (10-2)^2} = 10 \text{ ohm}, \quad I_s = 40/10 = 4 \text{ A.}$$

Step 3:

$$I_p = I_s/3 = 4/3 \text{ A}, \quad \cos\phi = R/Z = 6/10 = 0.6.$$

Step 4: Real power is

$$I_s^2 R = 16(6) = 96 \text{ W} = 120(4/3)(0.6),$$

checking the primary result.

Worked derivation:

$$S_p = S_s = 160 \text{ VA}, \quad P_p = P_s = 96 \text{ W}, \quad P/S = 0.6.$$

Why: Ideal transformation changes voltage and current levels but does not remove the load's phase difference. Multiplying primary voltage by current gives apparent power; the reactive secondary still makes the source power factor less than one.

Why not the others: (A) uses the secondary current at the primary. (B) assumes an ideal transformer makes every load resistive. (C) reverses current scaling and uses the reactance fraction as power factor.

Key point: Carry both magnitude transformation and load phase through an ideal transformer.

Q58 / A Spherical geometry produces two strict escape-angle branches for an internal source

Answer: (A) The two radial cones, $[0, \theta) \cup (\pi - \theta, \pi]$.

Given: A straight internal ray reaches a spherical boundary; OP=3R/4 and the internal refractive index is 2.

Formula/Principle: Express incidence through the ray's perpendicular distance from the centre, then apply the critical-angle condition.

Step 1: The ray line's perpendicular distance from O is

$$b = OP \sin\alpha = (3R/4)\sin\alpha.$$

Step 2: At the surface, the normal is a radius, so

$$\sin i = b/R = (3/4)\sin\alpha.$$

Step 3: The critical sine is $1/2$. Strict subcritical incidence requires

$$\sin\alpha < 2/3.$$

Step 4: On 0 through pi this means

$$0 \le \alpha < \theta \quad \text{or} \quad \pi - \theta < \alpha \le \pi.$$

Worked derivation:

$$i < i_c \iff (3/4)\sin\alpha < 1/2 \iff \alpha \in [0,\theta) \cup (\pi-\theta,\pi].$$

Why: Incidence at a curved surface is measured against the local radius, not the original OP direction. A ray initially directed inward can reach the opposite side with equally small incidence, so both near-outward and near-inward cones must be included. Critical rays are explicitly excluded; no equality endpoints are silently assigned to either behaviour.

Why not the others: (B) omits inward-directed rays escaping at the far side. (C) gives the complementary high-incidence directions. (D) ignores total internal reflection for more tangential directions.

Key point: For a curved boundary, relate the ray line's impact distance to the local surface normal before applying Snell's law.

Q59 / C Piecewise displacement-flux area gives equal magnetic fields at two different radii

Answer: (C) Equal magnetic fields with no gap conduction current.

Given: The idealised uniform electric-field region has radius a, and the gap is insulating.

Formula/Principle: $B(2\pi r) = \mu_0\varepsilon_0\, d\Phi_E/dt$. The enclosed field area changes at r=a.

Step 1: For r<a,

$$B(r) = \mu_0\varepsilon_0(r/2)\dot{E}.$$

Thus $B(a/2) = \mu_0\varepsilon_0 a\dot{E}/4$.

Step 2: For r>a, the enclosed flux uses area pi a squared:

$$B(r) = \mu_0\varepsilon_0\frac{a^2}{2r}\dot{E}.$$

Thus $B(2a) = \mu_0\varepsilon_0 a\dot{E}/4$.

Step 3: The values are equal even though one branch rises with r and the other falls as 1/r.

Worked derivation:

$$B(a/2)/B(2a) = 1, \qquad I_{\text{conduction,gap}} = 0.$$

Why: A changing electric flux contributes to magnetic circulation without transporting charge through the dielectric gap. Inside the plate radius only part of that flux is enclosed; outside, its entire value is enclosed while the loop circumference keeps growing. Applying one radial formula on both sides would give the wrong ratio.

Why not the others: (A) misuses the outer branch inside and invents conduction through the gap. (B) extends the inner branch outside and denies the changing-flux term. (D) omits displacement current from Maxwell's law.

Key point: Use the correct enclosed electric-flux area on either side of a charging capacitor's plate radius.

Q60 / B Efficiency change masks activity decay while background stays fixed

Answer: (B) 100 counts/min.

Given: Initial source-only signal is 80 counts/min and background is 20 counts/min; efficiency doubles after one half-life.

Formula/Principle: Detected source signal is proportional to efficiency times activity; background is added separately.

Step 1: Decay alone halves source activity, so at unchanged efficiency the signal would be

$$80/2 = 40 \text{ counts/min}.$$

Step 2: Doubling efficiency gives

$$S_f = 2(40) = 80 \text{ counts/min}.$$

Step 3:

$$C_f = S_f + 20 = 100 \text{ counts/min}.$$

Worked derivation:

$$C_f = 2\left(80 \cdot 2^{-1}\right) + 20 = 100 \text{ counts/min}.$$

Why: The detector can report the same mean rate even though half the radioactive nuclei have decayed. Instrument efficiency changes the fraction of emissions recorded, while the background is specified as unchanged. These are expected rates, not a claim that random observed counts are identical in every minute.

Why not the others: (A) accounts for decay but ignores the efficiency change. (C) doubles the original source signal without decay. (D) halves the total including a background that does not decay.

Key point: Separate activity, detection efficiency and background before interpreting a count-rate change.